

A Machine Learning Attack on Illegal Trading*

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Abstract

We design an adaptive framework for the detection of illegal trading behavior. Its key component is an extension of a pattern recognition tool, originating from the field of signal processing and adapted to modern electronic systems of securities trading. The new method combines the flexibility of dynamic time warping with contemporary approaches from extreme value theory to explore large-scale transaction data and accurately identify illegal trading patterns. Importantly, our method does not need access to any confirmed illegal transactions for training. We use a high-frequency order book dataset provided by an international investment firm to show that the method achieves remarkable improvements over alternative approaches in the identification of suspected illegal insider trading cases.

JEL Codes: G14, G18, C55

Key Words: illegal trading, insider trading, market manipulation, market micro-structure, order book, machine learning

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1 Introduction

Illegal trading practices undermine the integrity of financial markets. There is growing empirical evidence of illegal trading strategies, which include insider trading (Meulbroek, 1992; Kacperczyk and Pagnotta, 2019), closing price manipulation (Comerton-Forde and Putniņš, 2011) and spoofing (Lee et al., 2013). Successful market surveillance practices that are designed to detect illegal trading activity are known to improve the quality of financial markets (Fernandes and Ferreira, 2009; Cumming et al., 2011; Aitken et al., 2015). However, the precise identification of illegal trading activity poses a difficult problem because confirmed cases of illegal trading are rare when compared to the volume of legitimate transactions (Cline and Posylnaya, 2019). Moreover, what constitutes a pattern of illegal trading activity depends on the context and is generally not well defined, partly due to the need to avoid regulatory arbitrage (Putniņš, 2012). From a detection standpoint, these issues impede the application of classical supervised statistical learning techniques suited for balanced data and well identified patterns of illegal trading.

Our basic premise is that illegal trading activity differs substantially from patterns of historical legitimate trading activity at a broker. Hence, extreme abnormality of broker account trading activity is the key condition that constitutes a relevant alert in an effective market surveillance system. In this study, we combine the Nearest Neighbor Dynamic Time Warping (NN DTW) pattern recognition algorithm with extreme value theory to formulate an adaptive surveillance model designed to detect illegal insider trading activity¹. The design of our surveillance model is based upon time series anomaly detection techniques and leverages high-frequency broker-specific order book information to identify sequences of exceptionally abnormal trading activity by the broker's accounts. Importantly, our model does not require access to historical examples of illegal transactions for training and is also non-parametric in the sense that we make no assumptions on the data generating process or the structure of insider trading. We demonstrate that our anomaly detection approach identifies a greater proportion of all insider transactions while achieving a significantly lower false positive alert rate than alternative approaches from the literature.

We are particularly interested in studying insider trading activity because this behaviour, fundamentally underpinned by asymmetric information, is pervasive across financial markets. Moreover, insider trading has been linked to reduced investment and, as a consequence, to an inefficient allocation of capital and risk sharing across an economy (see, e.g., Manove, 1989; Ausubel, 1990; Leland, 1992; DeMarzo et al., 1998). Our proposed methodology is particularly relevant to brokers, exchanges and regulators because identifying insider trading practices is part of the mandatory transaction monitoring tasks faced by these institutions. The last decade has seen increasing ex-

¹For brevity, we refer to “illegal insider trading” as “insider trading” henceforth unless otherwise specified.

expectations from regulators to develop more comprehensive compliance and regulatory technology practices. As such, our study contributes to the advancement of regulatory market surveillance technology.

Traditional detection approaches assume that insider trading moves prices because the insider's private information is revealed to the market through the trading process. Based upon this assumption, insider trading can be detected by studying abnormal stock returns before informational events (see, e.g., Meulbroek, 1992). This idea has previously been used by regulators (Dubow and Monteiro, 2006; ASIC, 2019). However, we demonstrate that this abnormal return based detection approach does not adequately identify the presence of insider trading for the cases we investigate. In practice, individual brokers adhere to regulatory requirements by using rule-based market surveillance systems to monitor the trading activity of client and principle accounts for suspected illegal trading activities. These industry market surveillance systems struggle to adapt to dynamic market conditions and diverse patterns of normal trading behavior, thereby resulting in an exceptionally high rate of false positive alerts.

We focus on engineering features that monitor the trading behaviour of individual market participants, consistent with theoretical models that emphasize the strategic trading patterns of illegal insiders (see, e.g., Glosten and Milgrom, 1985; DeMarzo et al., 1998). An anomaly score is constructed by computing a measure of the distance based similarity to the nearest neighboring sequence in a historical set of legitimate broker account trading sequences. Fundamentally, our detection framework is based upon the fact that the signatures associated with insider trading are known to be markedly different from those associated with legitimate trading activity (Meulbroek, 1992; Shkilko, 2019; Kacperczyk and Pagnotta, 2019).

Trades with similar characteristics in two broker account trading sequences may be executed in a different order, for instance, due to time-varying liquidity and volatility in the underlying asset and/or as a result of a conscious attempt to conceal proprietary trading strategies. This means that traditional measures of statistical distance that rely on a linear alignment between trades in the feature space, such as the ℓ_p -norm², will incorrectly measure similarity. An incorrect measure of similarity would distort the estimated distribution of anomaly scores associated with legitimate trading activity, and hence our ability to precisely detect insider trading.

We address this challenge by applying the DTW algorithm, which is a pattern recognition algorithm known to perform exceptionally well in time series classification and clustering tasks (see, e.g., Ding et al., 2008; Dau et al., 2018). DTW finds the optimal non-linear alignment between similar trades that provide the best match for two trading sequences by realistically shrinking and stretching different regions of the sequences. An elastic distance measure, such as DTW, will be

²The ℓ_p -norm of an n -vector x is defined as $\|x\|_p = (\sum_{i=1}^n |x_i|^p)^{1/p}$

able to recover a more accurate estimate of similarity for time series that are misaligned. To the best of our knowledge, while the DTW algorithm has been widely used in the fields of computer science and engineering, its application to high-frequency financial transactions data is novel.

The upper tail distribution of historical anomaly scores is used to set a threshold that defines the boundary between legitimate and potentially illegal trading activity. We approximate only the upper tail of the anomaly score distribution using methods from the field of extreme value theory. This approach is asymptotically valid and permits inference beyond the range of observed data. Since extreme value theory is naturally used to describe the occurrence of rare events, the theory is particularly relevant to anomaly detection problems. We implement a sequential goodness-of-fit testing procedure with automated false discovery rate control within the extreme value modeling process. Our threshold estimation procedure requires no manual intervention by the surveillance analyst, as would be traditionally needed in applied extreme value modeling. The threshold is re-estimated daily so that detection rapidly incorporates changes in the distribution of legitimate broker account activity.

To evaluate our detection model we use second-by-second transaction data which covers 14 suspicious trading reports pertaining to alleged insider trading cases investigated by a global investment firm³, whom we have partnered with in this study. The 14 suspected insider trading cases originated in various prominent Asian equity markets over a four year period. The validation sample contains a total of 294 trading days of broker transaction activity across the 14 cases. In aggregate, our experiments analyze approximately 252,000 broker transactions and over 5,000 suspected insider transactions.

To the best of our knowledge, Olmo et al. (2011) and Park and Lee (2010) are the key studies that propose principled time series methods to detect insider trading.⁴ Both rely on modeling the stock return time series using traditional financial econometric methods. Olmo et al. (2011) test for the presence of structural breaks in the idiosyncratic component of stock returns using an Extended Capital Asset Pricing Model. This method was designed to detect long-lived insider trading over several months ahead of corporate announcements or unexpected news releases and therefore does not closely align with how broker market surveillance is conducted in practice. Park and Lee (2010) fit an ARMA(1,1) model to the intraday stock return time series and design three insider trading detection algorithms based upon the estimated model coefficients. We take this ARMA(1,1) model as the benchmark approach.

Using a time series cross validation testing procedure we demonstrate that our NN DTW model

³The investment firm acts as a broker-dealer across a large number of international equity markets.

⁴Islam et al. (2018) also offer a detection method but it is unclear from the paper whether the authors undertake a supervised or unsupervised approach and hence whether their approach is relevant for our setting.

identifies 90% of all suspected insider transactions in the validation sample. In contrast, the best ARMA(1,1) detection algorithm identifies around 60% of suspected illegal insider transactions. Moreover, our NN DTW model achieves an average false positive alert rate of 30%, while more than 71% of all transactions flagged as illegal by the ARMA(1,1) model were false positives. Hence, our surveillance model minimizes regulatory risk at the broker. We also show that our model outperforms an Ensemble Gaussian Mixture Model (EGMM), an isolation Forest (iForest) and a One-Class Support Vector Machine (OCSVM). In particular, the EGMM and the iForest are known to be powerful anomaly detection algorithms (Emmott et al., 2015). Our NN DTW model has the highest probability of assigning a large anomaly score to a randomly selected sequence of suspected insider trading activity and hence the highest probability of correct classification. We further demonstrate that our NN DTW model is able to achieve a dramatically lower false alarm rate than the other algorithms in a simulation exercise.

First and foremost, we contribute to literature that designs statistical models to detect illegal trading activity. Prior literature has applied time series econometric models (see, e.g., Olmo et al., 2011; Park and Lee, 2010) and supervised machine learning models to detect insider trading and manipulative trading activity more broadly. For instance, Diaz et al. (2011) use decision trees to identify patterns of stock market manipulations for a sample of eight manipulated stocks in 2003. The authors detect anomalies via a clustering technique, since the SEC does not disclose the exact dates of illegal trading activity. Using the same dataset and the same activity detection method, Golmohammadi et al. (2014) further examined the performance of random forests, Neural Networks, Naive Bayes, Support Vector Machines and the K-Nearest Neighbours supervised learning models. Finally, Deng et al. (2019) train gradient boosted decision trees to detect insider trading in the Chinese stock market.

We emphasize that these supervised approaches are generally not applicable to individual brokers, since the broker often does not have access to a sufficiently representative database of historical illegal trading activity to train a supervised learning model. For this reason, our approach is based upon the application of an one-class machine learning algorithm. Moreover, the aforementioned studies do not explicitly model the time series dimension of the data. A vast literature has shown that informed traders, of which illegal insiders are a subset, engage in multiple rounds of trading based on their private information (see, e.g., Glosten and Milgrom, 1985; He and Wang, 1995). Therefore, our detection model is explicitly designed to be applied to time series sequences of trading activity. Because our model achieves remarkable improvements over previously proposed approaches it could be used as a benchmark for future studies. We also provide new insights into insider trading by using a proprietary dataset and demonstrate the capabilities of different machine learning based anomaly detection models.

Aside from the machine learning dimension, our study contributes to the literature examining the behavior of illegal insider traders and factors affecting its prevalence and severity (see, e.g., Aitken et al., 2015; Cline and Posylnaya, 2019). In particular, Meulbroek (1992) and Kacperczyk and Pagnotta (2019) find that the presence of illegal insiders are related to heightened liquidity and volatility in the U.S. equity market. We draw upon this literature to design our surveillance model. Additionally, we provide new insights into how these findings apply across the largest Asian equity markets.

Our study is also related to the broader literature on data driven financial fraud detection. While the detection of credit card fraud (Bolton and Hand, 2002), financial statement fraud (Spathis et al., 2002) and insurance fraud (Derrig, 2002) have received significant attention within the literature, there are comparatively few studies that design detection systems for fraudulent and illegal activity in securities markets (Ngai et al., 2011). Our work addresses this gap in the literature. Moreover, we differ from much of the traditional financial fraud literature that uses supervised learning methods by framing the detection of insider trading as a one-class machine learning problem.

Finally, our anomaly detection approach is related to literature studying illegal trading activity in cryptocurrency markets. Consistent with our approach, Cong et al. (2021) and Amiram et al. (2020) demonstrate the utility of anomaly detection methods to identify instances of illegal trading. This work applies Benford's Law⁵ and the E-Divisive with Medians technique to identify change points. However, our anomaly detection approach is different, since we are designing a method to identify multivariate sequences of anomalous transactions that is based upon a measure of statistical distance, namely DTW. Nonetheless, we are optimistic that our approach could also be used to detect sequences of illegal trading activity in other asset classes which are traded on exchanges that use the limit order book mechanism.

The remainder of the paper is structured as follows. Section 2 describes the operations of an equity market surveillance division in a brokerage firm, the architecture of existing rule-based expert market surveillance systems and the limitations of these systems. Section 3 describes our data and features. Section 4 presents our anomaly detection method and the NN DTW algorithm. Section 5 details the component of our model which is based on extreme value theory. The performance of our detection model is evaluated in Section 6. Section 7 concludes.

⁵Benford's Law is an observation that in many naturally occurring sets of numerical data, the expected frequency of the first significant digit is likely to be small. Deviations from Benford's Law can be used to identify anomalous patterns in large datasets. For example, Benford's Law can be applied to identify the presence of an anomalous trade volume of a transaction which may be indicative of illegal wash trading.

2 Broker Market Surveillance

Market surveillance divisions use automated surveillance systems to generate alerts for instances of suspected illegal activity. These alerts are escalated to surveillance analysts who review the flagged activity to either close the alert as a false positive or conduct a detailed investigation. An important step in the alert review process requires the surveillance analyst to assess whether or not the flagged broker account traded in an unusual or anomalous manner given their recent trading history in the stock and/or the recent trading history of other broker accounts. It is important that the Type 1 error (false positive) rate of a surveillance system is kept as low as reasonably possible to ensure human analysts are not overwhelmed by false alerts. Moreover, since brokers may be subject to significant financial penalties if surveillance systems fail to adequately detect illegal trading activity, a market surveillance system must also be able to identify a large fraction of all illegal trading activity to ensure compliance with regulatory standards.

Currently, the majority of broker compliance divisions rely on rule-based expert surveillance systems to detect illegal trading activity⁶. These expert systems embed knowledge of illegal trading activity via a pre-set dictionary of human defined rules and parameters which specify the structure of known illegal trading scenarios. Each parameter in a scenario is associated with an individual threshold. These thresholds are largely time-invariant and are typically applied globally across a population of stocks and broker accounts being monitored. Alerts are generated using conditional logic which compares the values of the scenario parameters with their corresponding thresholds to determine if an alert should be triggered. Rule-based expert surveillance systems implicitly assume that the structure of illegal trading activity is known *ex-ante* and can be well defined by a series of conditional logic statements. See Golmohammadi et al. (2014) for a complementary description of the rule based experts systems that are used in practice.

The complex nature of modern financial markets means that an accurate rule-based representation of all possible modes of illegal trading activity is infeasible. This problem is compounded by the relative rarity of prosecuted trading behavior, meaning that domain experts have only a small number of cases on-hand to design and calibrate alert scenarios and thresholds. As such, rule-based systems suffer from an incomplete monitoring problem, whereby instances of illegal trading activity which do not strictly match the conditional logic in alert scenarios are unlikely to be detected.

The time-invariant global thresholds used in current rule-based expert surveillance systems lead to a high Type 1 error rate, since they cannot rapidly adapt to changing market conditions. This, in turn, can lead to complacency in the manual alert review process, a problem which is sometimes referred to as the base rate fallacy (see, e.g., Axelsson, 2000). Moreover, the alerts generated by

⁶The NASDAQ SMARTS and ONETICK systems are examples of rule-based expert market surveillance systems.

the existing systems provide little information about the abnormality of trading patterns⁷, despite this being a critical component of the alert review process. The market surveillance model that we design in this study seeks to overcome these limitations.

3 Broker Transaction Data and Financial Feature Engineering

Order-driven exchanges operating electronic limit order books use a continuous double auction mechanism to match buy/sell orders stored in bid/ask queues. Four events occur in order-driven markets, namely order submissions, order cancellations, order amendments and trade executions. Given our goal of insider trading detection, we are concerned primarily with trade executions and best bid/ask submissions.

Typically a broker's equity market surveillance department analyzes a semi-identified limit order book, that is a limit order book in which all executed trades and best bid/ask quotes are observed but only a subset of these trades and quotes is associated with the named client/principle trading accounts of the broker. The trading accounts of other brokers are anonymous and cannot be distinguished from one another. Each transaction in the data is associated with (i) a price, (ii) a volume, (iii) a time rounded to the nearest second⁸, (iv) an account identifier which is either the broker's client account code if that broker's account was on one side of the trade or blank otherwise, (v) the prevailing best bid and ask prices and (vi) qualitative trade flags (for example, whether or not the trade was a short-sell or buyer/seller initiated). On each day transactions are sequenced in ascending order according to their timestamp, rounded to the nearest second.

In general, the calendar time duration between consecutive trades is not the same as trades are irregularly spaced in time. In this study we consider sequences indexed by the order of transaction arrival rather than calendar time. Hence, time is incremented each time a new trade arrives and refers to the relative position of a transaction within the sequence. This approach is known as transaction time sampling in the literature (Hautsch and Podolskij, 2013; Hasbrouck, 1991). Instead, we capture the calendar time properties of trading activity through our feature engineering process.

Given the one-class nature of the insider trading detection problem, standard feature selection techniques, such as penalized regression models like Lasso, cannot be used to identify the optimal feature set for detection. Instead, we define a set of relevant microstructure variables by surveying the literature on market dynamics with asymmetric information and heterogeneous agents. We also

⁷we note that the identification of anomalous trading patterns by market surveillance systems serves a secondary function, which is to ensure that the occurrence of manual trading errors, the so called 'fat finger' errors, is minimized

⁸This was the minimum time increment available in our empirical tests so we use it to describe our data environment. Higher frequency data does not change our proposed detection model.

leverage the practical surveillance experience of an investment firm that provided the data used in our empirical experiments. Broadly speaking, the theory of markets with asymmetric information postulates that the trading activity of privately informed market participants alters the observable value of relevant market variables (Kacperczyk and Pagnotta, 2019). Hence, insider trading can be detected by identifying abnormal changes in a set of relevant market variables.

The first class of variables we consider are measures of trade volume and transaction frequency. Theoretically, trade volume is correlated with the degree of private information held by informed market participants (Easley and O’hara, 1987; He and Wang, 1995; Glosten and Milgrom, 1985). As features we compute the raw trade volume associated with each transaction, a measure of broker account trade volume which does not reflect liquidity changes that may be associated with exogenous market events, the volume which originates from short sell orders, the sum of trade volume for each broker account over 30-minute trade-by-trade moving windows, and a measure of the intraday broker account participation rate over the same windows.

Our second set of features measure liquidity provision and demand at the time of broker account transactions, since theoretical models of liquidity closely consider the behaviour of informed traders (see, e.g., Glosten and Milgrom, 1985; Kaniel and Liu, 2006; Lim and Coggins, 2005). We compute the immediate price impact and the permanent price impact of a transaction (also known as the adverse selection spread component), and a measure of trade duration which is adjusted for intraday periodicity to capture the liquidity dimension of broker account trading activity. Finally, we compute a measure of intraday realized volatility over 30-minute trade-by-trade moving windows using grids of 1,800 transaction prices. Our volatility feature is the equal-weighted average of 40 microstructure noise robust realized volatility estimators (Patton and Sheppard, 2009). We provide mathematical definitions and further motivation for each of our features in the online supplementary materials file.

Feature engineering is performed on the entire semi-identified order book data. Then, for each stock, we extract all intraday trading sequences associated with the identifiable broker accounts from the entire dataset. For each broker account on each day, we construct trade-by-trade sliding windows of m consecutive trades in a given stock. These windows form our intraday trading sequences. Any trading sequence shorter than m trades is up-sampled via linear interpolation to length m . The set of all trading sequences for a given stock, account and day comprises what we call a broker account trading activity.

Formally, each trade is characterized by a p -dimensional feature vector $\mathbf{v}_j$. Then, an intraday broker account sequence is associated with an $m \times p$ data matrix, which we denote by $\mathbf{V}_{id} = \{\mathbf{v}_j, j = 1 \dots, m\}$, where i indexes the broker account trading sequences. The information related to all broker account trading activity can be represented by the set $\{\mathbf{V}_{id}, i = 1, \dots, N_d\}_{d=1}^D$, where

N_d is the number of intraday trading sequences on day d , and D is the number of days in the sample.

4 Methodology

Two aspects of the equity market microstructure guide how we implement the time series anomaly detection method for insider trading detection. First, order-driven equity markets operate separate limit order books for individual securities. The idiosyncratic characteristics of individual stocks imply that each order book will comprise a different set of normal trading activity. We argue that a stock specific modelling approach reduces Type 1 errors and delivers improved classification performance. The rationale is that the thresholds which define the boundary between normal and anomalous trading activity can be learned more accurately when using only stock specific information. Moreover, this approach aligns well with the structure of existing expert market surveillance systems, enhancing synergies between our bottom-up approach and the existing top down approach used in practice.

Second, insider trading takes place on only one side of the market, as the insider seeks to profit from an anticipated price movement. Therefore, we construct separate anomaly detection models for the bid and ask sides of the order book. This approach may also be beneficial in cases where the insider attempts to disguise their illegal trades by placing seemingly legitimate trades on the opposite side of the order book (see, e.g., Back and Baruch, 2004). In what follows, unless otherwise stated, we omit the subscripts representing individual stocks, brokers and the sides of the order book.

4.1 Anomaly Detection Framework

Let $\{\mathbf{V}_{id}, i = 1, \dots, N_d\}_{d=1}^D$ denote the available information set. We split the information set into two subsets: the training dataset of legitimate trading activity $\mathbf{X} = \{\mathbf{V}_{id} : i = 1, \dots, N_d, d \in I_1\}$, where $I_1 \subset \{1, \dots, D\}$, and the testing dataset $\mathbf{Y} = \{\mathbf{V}_{id} : i = 1, \dots, N_d, d \in I_2\}$, where $I_2 = I_1^c$ and $\min(I_2) > \max(I_1)$. All activity in the training dataset with days indexed by I_1 is designated as legitimate trading activity because we have no cases of verifiable manipulation for those days.

Our detection framework is based on a comparison of $\mathbf{Y}$ to $\mathbf{X}$. We define the anomaly scoring function as $f(\mathbf{X}, \mathbf{Y}, \theta)$, where θ is a set of hyper-parameters. This function generates anomaly scores for all intraday sequences in $\mathbf{Y}$ based on information about trading activity in $\mathbf{X}$ using the NN DTW algorithm. The main idea is to use $\mathbf{X}$ to learn what trading behavior is considered normal and then detect sequences in $\mathbf{Y}$ with trading patterns which do not reasonably conform to the patterns in $\mathbf{X}$. We achieve this in the following two step procedure.

First, the anomaly scoring function is applied to sequences of legitimate trading activity to generate a set of training sample anomaly scores $\vartheta^{(X)} = f(\mathbf{X}, \mathbf{X}, \theta) = \{\vartheta_{id}, i = 1, \dots, N_d : d \in I_1\}$. Let F denote the distribution of $\vartheta^{(X)}$, which is a continuous distribution. We use a sample of upper order statistics from $\vartheta^{(X)}$ to estimate the parameters of a generalized Pareto extreme value distribution which serves as an asymptotically valid approximation of the upper tail of F . An anomaly score threshold is set as the conditional tail expectation $\tau^{(\alpha)}$ at confidence level α , using this approximation.

Second, we apply the anomaly scoring function to the testing data. This generates a set of testing sample anomaly scores $\vartheta^{(Y)} = f(\mathbf{X}, \mathbf{Y}, \theta) = \{\vartheta_{id}, i = 1, \dots, N_d : d \in I_2\}$ for each trading sequence in $\mathbf{Y}$. Using the anomaly score threshold $\tau^{(\alpha)}$, the class label vector is then $\hat{\mathbf{y}} = -\text{sign}(\vartheta^{(Y)} - \tau^{(\alpha)})$. A sequence is flagged as anomalous and thus representative of potential insider trading activity, if the anomaly score exceeds the threshold (i.e. if $\hat{y}_{id} = -1$).

4.2 Basics of Dynamic Time Warping

The ℓ_p -norm is a common measure of statistical distance. When applied to time series data the ℓ_p -norm enforces a linear alignment between observations in two sequences. However, time-varying market conditions, such as liquidity or volatility, and/or because of conscious attempts by traders to conceal trading patterns by randomly shuffling the order of a small number of transactions may cause regions of similar transactions for two sequences of trading activity to become misaligned. To address this issue, we measure the statistical distance between sequences using DTW.

DTW is an elastic distance measure that finds the optimal alignment between two sequences by constructing a non-linear mapping between observations (Sakoe and Chiba, 1978). The basic idea is to warp, that is to stretch and shrink, various parts of a query sequence to find the best match to a reference sequence. In doing so, one can define a distance measure that is invariant to any misalignment in time between similar regions of the two sequences. The exceptional performance of DTW for time series similarity search tasks is well established in the literature (Ding et al., 2008; Bagnall et al., 2017; Dau et al., 2018).

The idea of non-linearly warping a sequence is easily seen in Figure 1, by comparing the alignment produced using DTW with the alignment produced using a standard ℓ_p -norm, such as the Euclidean norm. As an illustration, the first panel of Figure 1 plots two starlight curves for the same class of star which are misaligned in time⁹. Clearly, a Euclidean alignment does not account for the misalignment between the two starlight curves. However, using DTW we observe that the blue curve is warped so that similar regions of the two curves are aligned along the time axis.

⁹The starlight curve data is available at https://www.cs.ucr.edu/%7Eeamonn/time_series_data_2018/

The fourth panel of Figure 1 shows the two series after they have been synchronized using DTW. The two starlight curves are then shown to belong to the same star. Figure 1 demonstrates that measures which enforce lock-step one-to-one mappings, or even fixed linear mappings, between observations can considerably overstate the distance between sequences which are unpredictably misaligned in time.

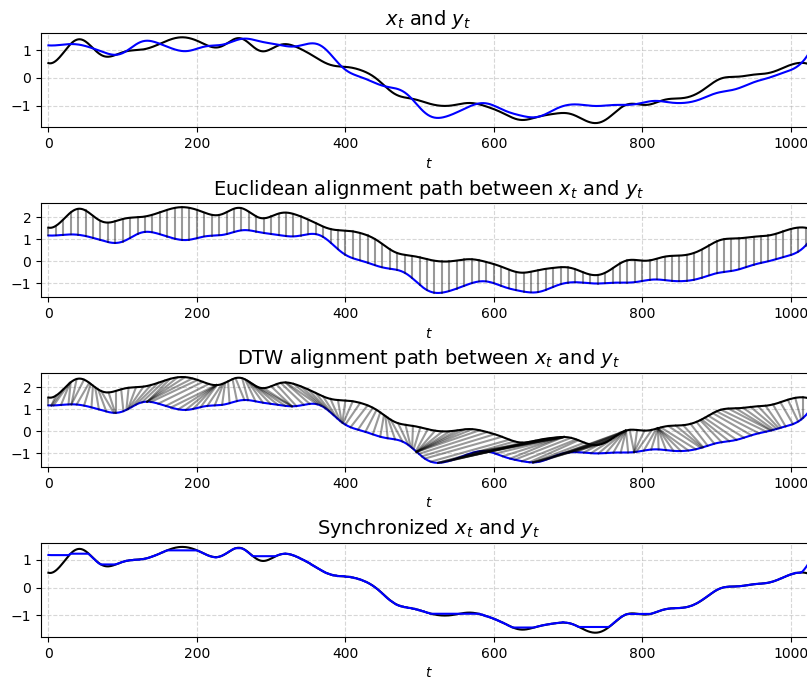


Figure 1: Comparison of Euclidean and DTW alignment between two starlight curves for an identical star.

4.3 Multivariate DTW

When time series are multivariate (vector valued) DTW can be generalized in one of two ways (see, e.g., Shokoohi-Yekta et al., 2017). First, one may construct feature-specific alignments by applying DTW to the time series for each feature in the multivariate sequences. The distances corresponding to each feature-specific alignment are then summed to compute the overall distance between the multivariate sequences. This approach is known as independent DTW since it assumes that the features carry no joint information so that characteristics of one feature do not affect the warping path of another. The alternative is to warp all dimensions simultaneously based upon a

single optimal alignment path. This is known as dependent DTW. For problems where the feature time series are “tightly coupled”, that is they change simultaneously as new observations arrive, dependent DTW is preferred. In our setting, a new trade mechanistically affects the values of all features simultaneously, giving rise to a high degree of dependence. We, therefore, implement the dependent version of multivariate DTW. All references to DTW herein refer to the dependent variant.

Broadly described, the DTW distance between two time series sequences is calculated in three steps. First, a so-called cost matrix that contains the Euclidean distances between all possible combinations of observations in the two time series is computed. Second, the optimal warping path that defines the alignment between observations is found using dynamic programming. This optimal warping path is found subject to constraints that ensure a realistic mapping between the two time series. Finally, the DTW distance is calculated as the square-root of the sum of distances along the optimal alignment path. Appendix A describes the DTW algorithm in more detail.

To visualize the mechanics of DTW, Figure 2 plots the cost matrix and cumulative cost matrix together with the optimal warping path (solid white line) obtained under DTW for the two starlight curves introduced in Figure 1. An example of the Warping Window Width constraint (See (10) in Appendix A), is plotted as the dashed lines around the diagonal of the cumulative cost matrix in panel (b). Intuitively, one can observe that the optimal warping path is the path of least resistance through a hilly surface where the values of $C(i, j)$ measure the hill heights.

4.4 The Nearest Neighbor Dynamic Time Warping Algorithm

The DTW distance is used as the statistical distance measure in a standard one-NN algorithm. We believe the one-NN algorithm is an attractive solution to the broker market surveillance problem for three reasons. First, a one-NN algorithm has no need for hyperparameter tuning, which can be exceptionally difficult in one-class learning problems. Second, a NN algorithm is non-parametric. Since no explicit definition of what constitutes insider trading activity is used to construct the anomaly score, it is possible that multiple patterns of insider trading behavior can be identified using this anomaly detection approach. This sets our detection model apart from other insider trading detection approaches in the literature and addresses the limitations of the existing market surveillance systems that rely upon conditional alert logic. Third, the instance-based learning property of NN algorithms can handle concept drift by allowing the training dataset and threshold to be updated at a high frequency with negligible model retraining costs. This property ensures regulatory compliance as the most recent trading information can be rapidly incorporated into the surveillance model.

In its basic form, the complexity of the NN DTW algorithm is considerable, generally inhibiting

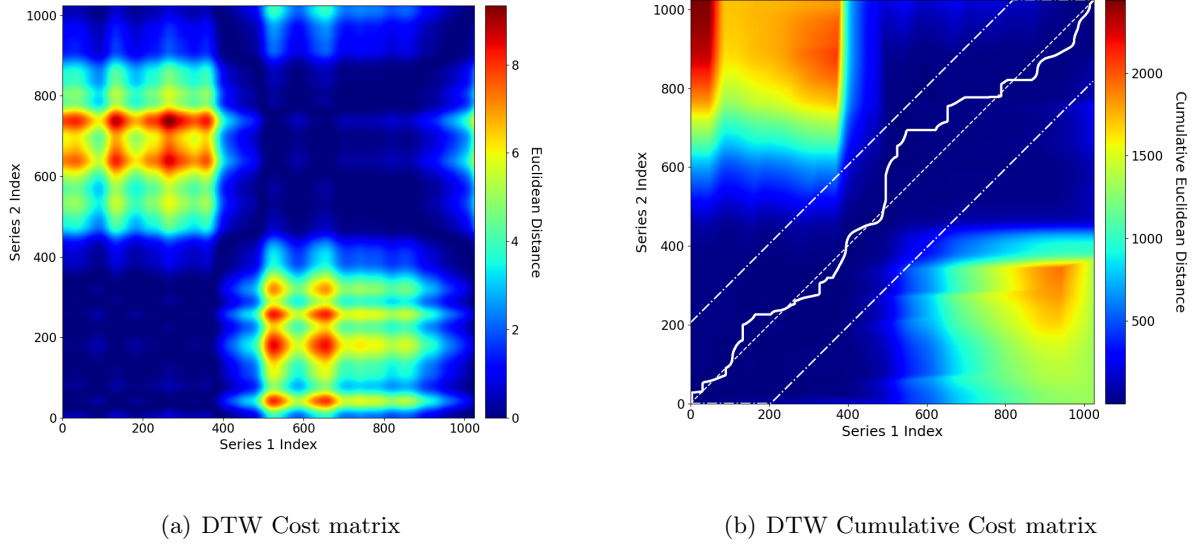


Figure 2: DTW cost matrices, optimal warping path and warping window width constraint for the alignment of the two starlight curves from Figure 1. The Euclidean distance is 8.645. The DTW distance is 3.665.

its application to large scale time series classification problems, such as financial market surveillance¹⁰. However, several studies have proposed alternative procedures to reduce the complexity of the NN DTW algorithm. Broadly speaking, these procedures use the constraints imposed upon the optimal warping path to prune nearest neighbor candidates. We use three lower bounding functions and two early abandoning techniques to prune nearest neighbor candidates. We describe each technique in Appendix B and give the pseudo code for the NN DTW algorithm. In our experiments these techniques reduce the number of required full DTW calculations by approximately 85%, resulting in a considerable reduction in computational complexity. Finally, we use the efficient cost matrix construction procedure of Silva et al. (2018) to reduce the search space of the optimal warping path by pruning regions of the DTW cost matrix.

5 Threshold Selection

We address the setting of a precise anomaly score threshold for one-class learning problems, using only information about the negative class. This threshold defines the largest anomaly score for

¹⁰Since DTW distance does not satisfy the Triangle Inequality, well known indexing techniques for metric spaces cannot be used to reduce complexity.

which a broker account sequence can be classified as normal. Typical non-parametric approaches to select an anomaly score threshold involve using quantiles of the empirical anomaly score distribution function, which is equivalent to selecting the anomaly score value that identifies the n largest scores as outliers. However, the estimation of high quantiles in finite samples can be subject to large variance. Moreover, only previously observed values of anomaly scores can be used as thresholds and estimation beyond the largest observed value is not possible when using the empirical distribution function. Parametric approaches typically involve assuming the entire range of anomaly scores follows a specific distribution, such as Gaussian (Ahmad et al., 2017). Strong parametric assumptions are subject to model misspecification risk, particularly so if the underlying anomaly scores are heavy tailed and this regularity is not reflected in the choice of distribution.

Instead, we rely on extreme value theory to construct an accurate representation of the upper tail of the distribution of training sample anomaly scores. Extreme value theory describes the asymptotic behavior of a stochastic process at unusually large or small values. The theory can be used to construct a model for the tails of an underlying distribution, without the need to make potentially inappropriate parametric assumptions over the entire support of the underlying distribution. We rely on the “Peaks-Over-Threshold” (POT) method¹¹ which models the behavior of observations above a high initial threshold value, denoted by u .

We are given a set of training sample anomaly scores $\vartheta^{(X)}$ computed by the NN DTW algorithm which are assumed to be i.i.d. The common but unknown distribution function of these scores is denoted by F . We can define an anomaly score threshold as the value in the support of F such that at most $1 - \alpha$ probability mass is above this value, for some sufficiently large α , since we are interested in identifying anomalous broker account sequences. Quantiles are natural here. To apply the POT method we fit the generalized Pareto distribution to the exceedances of the anomaly scores above the initial threshold value. We provide additional details regarding the foundations of POT extreme value theory in Appendix C. Let $\hat{\xi}$ and $\hat{\sigma}$ denote the estimated shape and scale parameters of the generalized Pareto distribution respectively. The POT method derives the following estimator of the tail α -quantile

$$\hat{Q}^{(\alpha)} = u + \frac{\hat{\sigma}}{\hat{\xi}} \left[\left(\frac{N}{N_u} (1 - \alpha) \right)^{-\hat{\xi}} - 1 \right], \quad \alpha > F(u) \quad (1)$$

where N denotes the number of anomaly scores in the training data and N_u is the number of

¹¹ An alternative is to use the “Block-Maxima” method, which segments observations that are measured over regular intervals into non-overlapping blocks and uses the largest order statistic in each block to develop a model for the tail distribution. However, there is no obvious choice of block size in our application. Moreover, the block-maxima method uses available data inefficiently by sampling only the largest order statistic in each block.

anomaly scores that exceed u .

The quantity $\hat{Q}^{(\alpha)}$ is akin to the so-called Value-at-Risk when the underlying data are negative log returns of financial assets. However the use of $\hat{Q}^{(\alpha)}$ as an anomaly score threshold leaves room for improvement, since information beyond the quantile $\hat{Q}^{(\alpha)}$ is disregarded. One implication, that sequences with anomaly scores which are marginally greater than $\hat{Q}^{(\alpha)}$ will be flagged as anomalous even though they are typically not much different from anomaly scores just below $\hat{Q}^{(\alpha)}$.

A more robust anomaly score threshold would consider the *average* behavior of the training sample anomaly scores exceeding $\hat{Q}^{(\alpha)}$, that is the behavior of $E[\vartheta_{id} - \hat{Q}^{(\alpha)} | \vartheta_{id} > \hat{Q}^{(\alpha)}]$. In the financial risk management literature this measure is typically referred to as the Expected Shortfall, while more broadly it is known as the Conditional Tail Expectation (CTE) or Superquantile. Subject to the assumption that $\xi < 1$, which ensures that the mean of the generalized Pareto distribution is finite¹², the CTE anomaly score threshold is

$$\hat{\tau}_\alpha = \frac{\hat{Q}^{(\alpha)} + \hat{\sigma} + \hat{\xi}(\hat{Q}^{(\alpha)} - u)}{1 - \hat{\xi}}, \quad \alpha > F(u), \quad \hat{\xi} < 1 \quad (2)$$

The CTE anomaly score threshold can be understood as the average anomaly score of the most unusual patterns of normal training activity. The level of confidence α defines where unusual training sample anomaly scores begin in terms of the quantiles of F . Hence, sequences in the testing sample are flagged as anomalous only if they are exceptionally different from historical normal trading activity. Unusual, but still legitimate trading activity, such as fund fire sales or large scale portfolio re-balancing, could be associated with large training sample anomaly scores. In these cases, F will likely have heavy tails if unusual but still legitimate trading events occurred during the training sample. The extreme value based anomaly score threshold accounts for this circumstance by accurately modeling the behavior of the tails of F when such tails are heavy. Since the construction of the anomaly score threshold is reminiscent of financial risk management practices we believe the choice of α is intuitive and would typically lie between 0.95 and 0.99. This means that the anomaly score threshold is set based upon the average behavior of the largest 5% (or less) of training sample anomaly scores.

The POT extreme value modeling approach described above relies on the assumption that observations are stationary and i.i.d. We use a sliding window of size m to define all possible combinations of intraday trading sequences for each broker account. As such, anomaly scores for accounts that execute a large share of daily trades may not be independent of one another. To eliminate this source of potential dependency we reduce the set of training sample anomaly scores

¹²In most practical applications it is unlikely that ξ exceeds one. In cases where $\xi > 1$ the quantile based anomaly score threshold $\hat{Q}^{(\alpha)}$ can be used instead.

used to estimate the parameters of the generalized Pareto distribution by retaining only scores for non-overlapping broker account sequences on any given day¹³. This reduced set of training sample anomaly scores is denoted by $\tilde{\vartheta}^{(X)}$.

The use of a generalized Pareto distribution requires the selection of a value for the initial threshold u , which yields a trade-off between bias and variance. To select the optimal u we take advantage of the automated sequential testing procedure of Bader et al. (2018). Specifically, we consider an ordered finite set of candidate thresholds $\tilde{u}_1 < \dots < \tilde{u}_j < \dots < \tilde{u}_n$ between the 70th and 94th quantile of the training sample anomaly scores in increments of one. Each candidate threshold is associated with a sample of exceedances $\tilde{Z}_j = \{\tilde{\vartheta}_{id} - \tilde{u}_j | \tilde{\vartheta}_{id} > \tilde{u}_j, i = 1 \dots N_d : d \in I_1\}$, $j = 1, \dots, n$. The j^{th} null hypothesis, $H_j^{(0)}$, states that $\tilde{Z}_j$ follows a generalized Pareto distribution. The Anderson-Darling goodness-of-fit test is applied to each set of exceedances with the corresponding threshold, starting with $(\tilde{u}_1, \tilde{Z}_1)$ and proceeding until a pair $(\tilde{u}_j, \tilde{Z}_j)$ results in an acceptance of the null hypothesis. We utilize the interpolated table of p-values provided by Bader et al. (2018).

The rejection of any null hypothesis $H_j^{(0)}$ implies that all prior null hypotheses for lower candidate thresholds have also been rejected. Since multiple hypotheses are effectively being tested there is a need to control the false discovery rate. Bader et al. (2018) propose to apply the recently developed ForwardStop rule of G'Sell et al. (2016), which was shown to control the false discovery rate at a level β in settings which require ordered hypothesis testing. After computing the p-values p_j , from the Anderson-Darling test for all candidate thresholds the index of the optimal threshold is

$$\hat{k} = \max\{k \in \{1, \dots, n\} : -\frac{1}{k} \sum_{j=1}^k \log(1 - p_j) \leq \beta\} \quad (3)$$

and the optimal threshold is $u = \tilde{u}_{\hat{k}}$. We set $\beta = 0.1$.

Therefore, our anomaly score threshold estimation procedure can be applied without needing to manually adjust parameters, as is traditionally the case when applying extreme value theory to data. Moreover, the anomaly score threshold can be made time-varying by simply re-estimating u , ξ and σ whenever the set of training sample anomaly scores is updated. Users are required to pre-select only one parameter, namely the quantile level α , to estimate the anomaly score threshold.

6 Experiments & Evaluation

The described model is evaluated using data relating to 14 suspicious trading reports filed for alleged insider trading by an investment firm who acts as a dealer-broker in a large number of

¹³The Ljung-Box tests of the null hypothesis that the reduced anomaly score sample is independently distributed found no evidence against the null.

equity markets. The 14 suspected insider trading cases originated in various prominent Asian equity markets. Each case involves suspected insider trading activity ahead of an unscheduled material information release. These information releases are diverse and encompass financial announcements as well as news reports of corporate scandals. In each case, a material price movement alert was triggered on the information release date and a skilled human analyst identified the pattern of insider trading by manually analyzing historical transaction data.

Semi-identified limit order book data for each case in the sample identifies all trades by the suspected insider account in conjunction with all trades by the client accounts. As such, our experiments assess the ability of an insider trading detection model to distinguish transactions by the suspected insider account from all other broker account transactions in the days preceding the information release. We use a validation sample containing all broker account transactions over a 21 trading day period (approximately one trading month) before the information release date which triggered the search for insider trading in each case. Hence, we analyze a total validation sample of 294 days of broker trading activity in our experiments. The training dataset is a rolling six-month historical sample of broker account transaction activity for each case. To ensure that the training dataset contains only normal legitimate transaction activity we remove any transactions associated with historical alerts produced by the existing rule-based expert market surveillance system, regardless of whether or not the alert was a false positive.

6.1 Summary Statistics

Table 1 provides an insight into the volume of trading activity for each suspected illegal insider account in our sample. The typical alleged illegal insider begins trading approximately four days before the information release date and executes a total of 421 trades based upon their private information. This corresponds to an average of 188 trades per day the suspicious account was active in the market. In aggregate, we study 5902 transactions associated with suspected insider trading from the suspicious trading reports. To the best of our knowledge, this is the largest sample of illegal insider trades studied to date within the literature¹⁴. The dollar value of trade volume per case varies between US\$0.5 million and US\$90 million, with the typical suspected insider trading US\$10.66 million ahead of the information release date. The trading activity statistics presented in Table 1 highlight the diversity of suspicious trading behavior.

Table 2 reports the cumulative abnormal returns over a 10-day pre-release period. This period covers all suspicious transaction activity in our sample. The information release date abnormal return is also reported. Our treatment of the pre-information release period cumulative abnormal

¹⁴For comparison, Kacperczyk and Pagnotta (2019) study 5,058 transactions associated with insider trading, of which 3392 are executed in U.S equity markets.

return follows the methodology used by regulators to measure the so-called "market cleanliness" (Dubow and Monteiro, 2006; ASIC, 2019). We use the market index relevant to each alleged insider trading case to compute daily abnormal stock returns. Table 2 therefore provides an insight into the effectiveness of traditional low frequency information leakage based insider trading detection methods that are used by regulators.

The information release day abnormal return $AR(0)$ exceeds 5% in absolute value in all but one case. The mean abnormal return on the information release day is approximately 11% in absolute value. Hence, the cases we study involve suspicious trading ahead of material single day price movements. In only two cases are the pre-release period cumulative abnormal returns $CAR(-10, -1)$ statistically significant and consistent with the direction of the inside information. Moreover, in approximately one-third of cases, cumulative abnormal returns in the pre-information release period are in the opposite direction to the information release day abnormal return. Hence, traditional insider trading detection approaches based upon the analysis of cumulative abnormal returns would have been ineffective in identifying the presence of suspected insider trading for 12 of the 14 cases.

Finally, Table 3 presents summary statistics for each feature used in the NN DTW detection model. Statistics are computed separately for all transactions in the first six-month training dataset and total 21-day validation sample. T-statistics are reported for differences in the mean of each feature between the training and validation samples. Both raw trade volume and short sell volume are higher in the validation sample compared to the training sample and the difference is statistically significant at the 1% level. Moreover, both mean and median broker account participation rates are higher in the validation sample. Since the suspicious cases in our sample all involve trading ahead of *unanticipated* material information, these elevated trade volume and participation rates are consistent with the presence of informed trading rather than uninformed speculation.

While the immediate price impact of transactions does not differ between the two samples, the adverse selection spread component (permanent price impact) is significantly higher in the validation sample. This suggests that some broker accounts can predict short term price movements ahead of unanticipated information releases. Importantly, this contrasts with the mean value of the adverse selection spread component in the six-month training sample, which demonstrates that the typical broker account is generally uninformed during normal times.

6.2 NN DTW Parameter Settings

Our NN DTW model requires the selection of two parameters. First, the warping window width parameter used in constraint (10) is set to $\varphi = 0.2$. This means that we allow DTW to search for an optimal alignment within a region of 20 trades. In standard classification problems this

parameter can be learned via cross-validation (Dau et al., 2018). However, this is not possible in one-class learning problems. We choose the value of $\varphi = 0.2$, based upon the results of Kurbalija et al. (2014) who find that the nearest neighbor assignments using the DTW distance remain approximately stable above 0.15 for univariate time series. We acknowledge that our choice of $\varphi = 0.2$ is unlikely to be optimal for all cases but generally works well in our experiments. One may therefore further improve the performance of NN DTW by judiciously setting the warping window width parameter. The second parameter to be set is the confidence level α , which is used to determine the anomaly score threshold. We choose a value of $\alpha = 0.95$, which is likely to be the lowest threshold a broker would consider using.

6.3 Validation Framework and Performance Criteria

We use the ARMA(1,1) model (Park and Lee, 2010) as our first benchmark because this method is designed for time series data and does not require access to historically prosecuted cases of insider trading when estimating model parameters. We implement each of the three detection algorithms proposed by (Park and Lee, 2010). As outlined in Section 2, brokers are required to monitor the trading activity of all client/principle accounts on each day before progressing to the next. Therefore, to emulate the structure of a real-world market surveillance system, we design a time series cross validation procedure to evaluate the detection performance of the NN DTW model and the ARMA(1,1) model of Park and Lee (2010). We provide details regarding the implementation of the ARMA(1,1) model in the online supplementary materials file¹⁵. This validation procedure is applied uniformly to each of the 14 suspicious trading cases. Figure 3 describes the time series cross validation procedure. Further details regarding the exact implementation of the cross validation procedure are given in Appendix D

Classification performance is evaluated using the concatenated validation sample which contains anomaly scores and class labels for all broker account sequences in each of the one-day testing periods from the time series cross-validation. The information retrieval performance of the NN DTW model and the ARMA(1,1) model of Park and Lee (2010) is summarised using the precision, recall and F1 scores. High precision scores ensure that the surveillance costs of a broker are minimized by reducing the volume of false positive alerts which must be investigated by surveillance analysts. High recall scores are of particular interest in market surveillance applications since financial penalties are imposed upon brokers who fail to detect illegal trading activity by their clients. A high F1 score indicates that both precision and recall scores are high. If a high recall score is achieved at the expense of a low precision score, and vice versa, then the utility of any

¹⁵The supplementary materials file is available at https://papers.ssrn.com/sol3/papers.cfm?abstract_id=3727753.

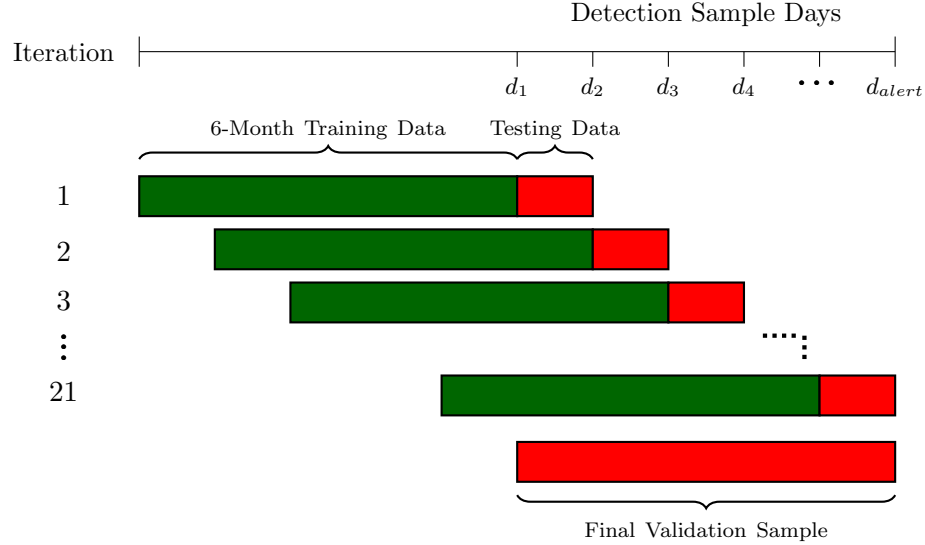


Figure 3: Time Series Cross Validation Procedure

insider trading detection model will be diminished and this will be reflected by a low F1 score. We compute individual precision, recall and F1 scores for each of the 14 cases in addition to the average precision, recall and F1 score across all 14 cases.

To examine threshold-independent classification performance we study the Area Under the Receiver Operating Characteristic (AUROC) curve and the Area Under the Precision Recall (AUPR) curve scores. Together, these two metrics describe how well a particular algorithm separates normal and insider trading activity, irrespective of the threshold. For each case of alleged insider trading 100 equi-spaced hypothetical thresholds in the interval $[\min(\{\vartheta_{id} | \hat{\mathbf{y}}_{id} = -1, i = 1, \dots, N_d : d \in I_2\}), \vartheta^{(Y)+}]$ are used to construct the ROC and PR curves. Vertically averaged ROC and PR curves are also computed. The area under each curve is computed using the trapezoidal rule.

To establish the statistical significance of differences in each of the aforementioned performance metrics we follow standard practice for the comparison of classifiers over multiple datasets (Demšar, 2006). Friedman tests are used to establish if each of the aforementioned performance metrics differs significantly between the NN DTW model and the three detection algorithms from the ARMA(1,1) model. Wilcoxon signed-rank tests are then used to assess the statistical significance of pairwise differences in the averages of each performance metric over all 14 cases.

6.4 Classification Performance Results

Table 4 reports the AUROC curve and AUPR curve statistics for our NN DTW procedure and the three ARMA(1,1) model detection algorithms. Figure 4 plots the vertically averaged Receiver Operating Characteristic (ROC) and Precision Recall (PR) curves to complement the numerical results in Table 4. The Friedman test statistics (p-values) are 27.0 (0.0) and 9.86 (0.02) for the averaged AUROC curve and AUPR curve metrics, respectively. Hence, the null hypothesis of equal classifier performance is strongly rejected.

First, our NN DTW model outperforms each of the ARMA(1,1) model detection algorithms for the AUROC curve score across all 14 cases. On average, our model achieves an AUROC curve score of 0.972, corresponding to an improvement of 0.329 on the highest performing detection algorithm from the ARMA(1,1) model. The difference in the average AUROC curve scores between our NN DTW model and each ARMA(1,1) detection algorithms is statistically significant at all conventional levels. Hence, our model has a higher probability of assigning larger anomaly scores to suspected insider transactions and therefore higher probability of correct classification.

Second, our NN DTW model also outperforms all three ARMA(1,1) detection algorithms for the AUPR curve statistic. The largest margin of difference in the average AUPR curve statistic is 0.191. The difference between the average AUPR curve scores for our model and each of the three ARMA(1,1) model detection algorithms is statistically significant at the 5% level. Figure 4 demonstrates that the true positive alert rate and the precision score of our NN DTW model are remarkably stable across all possible threshold choices. In summary, we demonstrate that the threshold independent classification performance of our model is superior to that of the ARMA(1,1) model.

Table 5 reports the precision, recall and F1 statistics for our NN DTW insider trading detection model and each of the three ARMA(1,1) model detection algorithms. The Friedman test statistics (p-values) for the precision, recall and F1 scores are 24.9 (0.00), 30.0 (0.00) and 34.27 (0.0) respectively. Hence, the null hypothesis of equal classifier performance is rejected for all three information retrieval statistics at all conventional levels of statistical significance. In all but three of the alleged insider trading cases (case 7, case 8 and case 14), our NN DTW model, outperforms all three of the ARMA(1,1) detection algorithms across all information retrieval statistics. Moreover, with respect to the F1 score, which measures the balanced precision and recall performance of a classifier, our model outperforms the ARMA(1,1) detection algorithms in all but one case (case 14).

We find that our NN DTW model identifies approximately 90% of all suspected illegal insider trades in the typical case over the 21-day validation sample. In contrast, the first and third ARMA(1,1) model detection algorithms only identify around 60% of all suspected insider trading

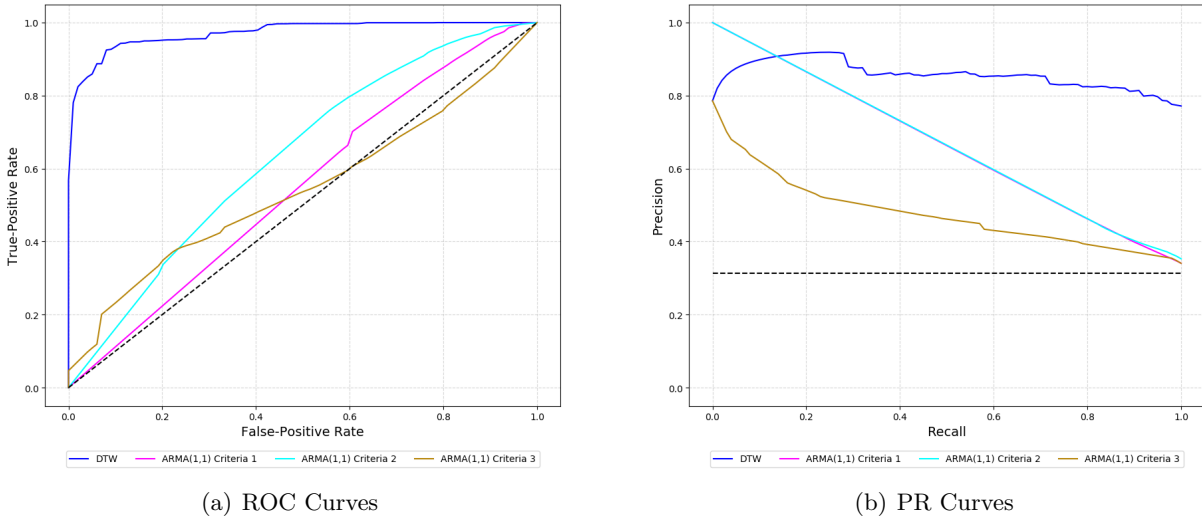


Figure 4: Vertically Averaged Receiver Operating Characteristic (ROC) and Precision Recall (PR) curves for the NN DTW and ARMA(1,1) illegal insider trading detection models

transactions. The second detection algorithm identifies only 5.4% of suspected illegal transactions. The differences in average recall between our NN DTW model and each of the three ARMA(1,1) model detection algorithms are statistically significant at the 1% level. As discussed in Section 2, brokers who fail to detect illegal trading activity may be subject to significant financial penalties. Hence, the average recall scores presented in Table 5 highlight the utility of our market surveillance for real-world market surveillance applications since this model minimizes regulatory risk at the broker.

Precision scores demonstrate that, on average, approximately 67% of all transactions identified by our NN DTW model belong to the suspected illegal insider account. This corresponds to a Type 1 error rate of 33%. None of the ARMA(1,1) detection algorithms achieve an average precision score greater than 0.29. The Type 1 error rate of the ARMA(1,1) model exceeds 71%. As such, the difference in average precision scores between our model and the ARMA(1,1) model is statistically significant at the 1% level. The low precision and high Type 1 error rate of all three ARMA(1,1) model detection algorithms suggests that in a real-world market surveillance system the volume of false positive alerts generated by the model could be unmanageable.

We find that the average F1 score of our NN DTW model significantly exceeds the F1 scores achieved by all three ARMA(1,1) detection algorithms and that these differences are statistically significant at the 1% level. The information retrieval performance of the ARMA(1,1) model suggests that the 10% significance threshold may not be optimal. Finally, we note that the total run time of

our model is approximately equal to that of the ARMA(1,1) model over the entire 21-day validation period. However, if the time taken to construct the initial set of training sample anomaly scores in the first iteration of the time series cross validation procedure is removed, our model is considerably faster than the ARMA(1,1) model. Additional details regarding this comparison can be found in the online supplementary materials file.

6.5 Point Anomaly Detection Model Benchmarks

An alternative to our time series modeling approach is to construct a model of normal broker account trading behavior based upon individual transactions (feature vectors), rather than sequences of transactions, thereby neglecting the temporal nature of the data. An advantage of this individual transaction based modeling approach is that several well studied point anomaly detection models can be readily applied. Point anomaly detection models refer to algorithms designed to identify anomalous data points rather than sequences (Chandola et al., 2009). Since many point anomaly detection models are implemented in open source machine learning software (Pedregosa et al., 2011), one may regard the approach as an out-of-the-box solution to the insider trading detection problem.

As a second benchmark for our NN DTW model we study how well point anomaly detection models perform in the insider trading detection problem. Guided by the review of Emmott et al. (2015) we select two state-of-the-art point anomaly detection models identified for general purpose applications, namely the EGMM and iForest algorithm of Liu et al. (2012). These two anomaly detection models are particularly powerful since they are ensemble learners. We also use a traditional point anomaly detection algorithm, namely the OCSVM with Gaussian kernel (Schölkopf et al., 2001). All the models use exactly the same set of features described in Section 3. The performance of each point anomaly detection benchmark model is evaluated using the time series cross-validation procedure described in Figure 3. The parameters of each model are re-estimated for each day in the validation sample as the rolling six-month training dataset is updated based upon the binary class labels assigned to broker account transactions in the previous iteration. Details regarding the implementation of the EGMM, iForest and OCSVM anomaly detection models can be found in the online supplementary materials file.

Table 6 presents the AUROC curve and AUPR curve statistics for the EGMM, iForest, OCSVM point anomaly detection models. For completeness, we re-tabulate the AUROC curve and AUPR curve statistics for our NN DTW detection model. Figure 5 plots the vertically averaged ROC and PR curves for our NN DTW detection model and the EGMM, iForest and OCSVM models. We find that our model outperforms the EGMM, iForest and OCSVM models with respect to both the mean AUROC curve and mean AUPR curve statistics. The difference in mean AUROC curve and

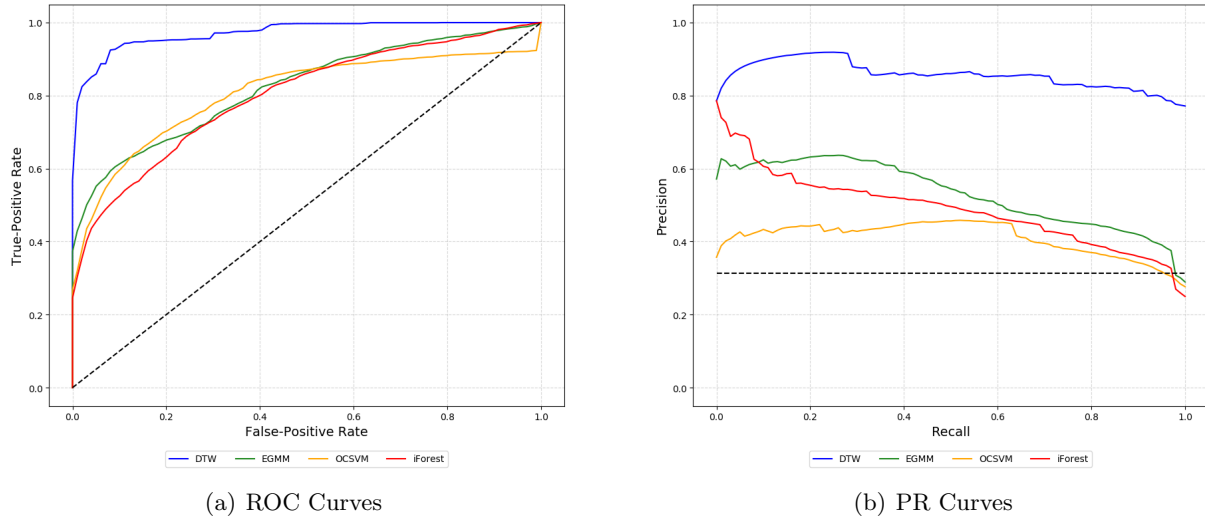


Figure 5: Vertically Averaged Receiver Operating Characteristic (ROC) and Precision Recall (PR) curves for NN DTW model and Ensemble Gaussian Mixture Model (EGMM), isolation Forest (iForest) and One-Class Support Vector Machine (OCSVM) models.

AUPR curve statistics for NN DTW model are statistically significant at the 1% level. The mean AUROC curve and AUPR curve scores for the EGMM and iForest point anomaly detection models are similar and outperform the OCSVM, likely since both are ensemble learners. Interestingly, we observe that the ARMA(1,1) detection algorithms 1 and 2 outperform the OCSVM, EGMM and iForest models in terms of the average area under the precision recall curve.

Table 7 reports the information retrieval statistics for EGMM, iForest and OCSVM models. Our NN DTW surveillance model continues to achieve substantially higher precision scores than any benchmark algorithm by a statistically significant margin. The Type 1 error rate for all three point anomaly detection models exceeds 57% in the typical case. Although this is an improvement over the Type 1 error rate of the ARMA(1,1) model, it is still considerably larger than the Type 1 error rate of our NN DTW model. The average recall score achieved by our model exceeds the mean recall scores of the EGMM and iForest models but is marginally below the mean recall score of OCSVM. However, the mean F1 scores for EGMM, iForest and OCSVM are all statistically significantly below the F1 score achieved by our NN DTW model. The F1 scores for the EGMM and OCSVM models demonstrate that these models achieve a high recall score at the expense of precision. This trade-off is generally unfavorable for practical market surveillance applications. Hence, we demonstrate that our NN DTW insider trading detection model also outperforms three point anomaly detection models, highlighting the utility of our time series detection approach.

6.6 Simulation Analysis

We simulate sequences of normal and anomalous trading activity to further study the detection performance of our NN DTW model compared to the OCSVM, EGMM and iForest benchmark detection models¹⁶. The procedure to simulate anomalous sequences that may be representative of insider trading activity is as follows. First, we randomly sample a broker account sequence from the training dataset of normal trading activity associated with case 12 on validation day d_1 . This randomly sampled broker account sequence is removed from the training dataset and the NN DTW, OCSVM, EGMM and iForest models are retrained using the revised training dataset. Let the variable $n.modified$ controls the number of transactions that are selected to be modified from the broker account sequence. Then, we replace the feature values for the set of features $\{trade\ volume, residual\ volume, moving\ volume\ sum\ and\ participation\ rate\}$ associated with these selected transactions with random values between the 99th and 100th quantiles in the training dataset. For example, simulations where $n.modified = 100$ represent cases where all transactions in the sequence are anomalous while simulations where $n.modified$ is less than 100 represent circumstances where the illegal insider “hides” anomalous transactions among normal transactions.

Finally, we add a degree of warping, controlled by the variable $warping$, to the modified broker account sequence using the algorithm in Dau et al. (2018). Each transaction in the simulated anomalous broker account sequence is assigned a true class label of -1 . These anomalous sequences simulate broker accounts that trade large volumes, since this is a common scenario observed in the actual insider trading cases studied above.

To simulate sequences of normal trading activity, we repeat the sampling process described above but modify a randomly chosen set of features for the sampled broker account sequence with values drawn at random from the entire empirical distribution of the training data for a random number of features. A simulated normal trading sequence is therefore a randomly constructed and warped permutation of the training dataset. Each transaction in the simulated normal broker account sequence is assigned a true class label of $+1$.

We consider values of the variable $n.modified \in \{50, 70, 90, 100\}$, values of the variable $warping \in \{30\%, 50\%\}$, and conduct 1000 replications of the simulation procedure to study classifier performance. Table 8 reports the classification rates for each detection model. When every transaction in a simulated broker account sequence is characterized by abnormally large volume features, all detection models are able to perfectly identify the presence of anomalous trading activity. In simulations where 90% of transactions are anomalous, our NN DTW model yields the highest True Negative classification Rate (TNR), that is 0.99 and 1 for the cases for $warping$ values of 30% and

¹⁶The code for the simulation exercise is available at <https://github.com/rjames6023/DTW-insider-detection>

50%, respectively. This result indicates that the model is able to correctly identify more insider trading activity than the benchmark detection algorithms. As the proportion of anomalous transactions drops to 70% the TNR of our NN DTW model remains competitive with the TNR of the benchmark algorithms. For the case where the proportion of anomalous transactions in the sequence drops to 50%, the OCSVM and the EGMM models have a marginally higher TNR than our NN DTW model. This result is intuitive, since the NN DTW model is a sequence based anomaly detection model while the OCSVM and EGMM models are point anomaly detection models. Our NN DTW model always has a higher TNR than the iForest model.

The results of the True Positive classification Rate (TPR) and False Negative classification Rate (FNR) highlight the superiority of the NN DTW model. Over all simulation scenarios, the FNR of our NN DTW model is negligible, demonstrating that the model raises very few false alarms in our simulations. This is an important result because the volume of false alerts generated by industry market surveillance systems is typically high. The FNR of the OCSVM and the EGMM models are high, suggesting that these models suffer from over-fitting and cannot identify randomly warped permutations of the training data. While the FNR of the iForest model is much lower than that of the OCSVM or EGMM models, it is still considerably larger than the rate associated with our NN DTW model. Reassuringly, the classification rates for our NN DTW model are nearly identical for both degrees of warping we consider.

7 Conclusion

Given the prevalence of asymmetric information in financial markets, the accurate and timely detection of insider trading is of paramount importance. In this study, we develop a financial market surveillance model for the early detection of insider trading. Our surveillance model addresses the limitations of existing approaches proposed in the literature and the shortcomings of industry standard market surveillance systems.

Our detection model relies on the application of the NN DTW algorithm to generate anomaly scores for discrete sequences of broker account trading activity. Using DTW distance, we overcome the difficulties associated with misalignment in transactions between two sequences of broker account trading activity, ensuring that similarity is accurately measured. We then construct an anomaly score threshold based upon the conditional tail expectation of anomaly scores in the training data using statistically principled methods from the field of extreme value theory.

Our empirical study analyzes 14 suspected insider trading cases from suspicious trading reports. This data pertains to a validation sample of 294 trading days of broker transaction data and over 5,000 suspected insider transactions. Our NN DTW model correctly identifies 90% of all suspected

illegal transactions in the validation sample and achieves a Type 1 error rate of 38% for the average case. We demonstrate that our detection model outperforms the ARMA(1,1) model of Park and Lee (2010) as well as three state-of-the-art anomaly detection techniques that could serve as out-of-the-box solutions to detect insider trading. Simulations further demonstrate that our NN DTW model raises considerably fewer false alarms than benchmark detection algorithms while still being able to competitively identify sequences of potentially illegal trading activity.

The surveillance model we develop could be used by regulators, exchanges and brokers, and is particularly suited to match the data constraints faced by individual brokers. The carefully designed anomaly scoring algorithm allows our surveillance model to achieve substantial improvements in detection performance over benchmark approaches, minimizing regulatory risk at the broker. We believe that our one-class machine learning approach can be used in conjunction with existing expert market surveillance systems to provide a holistic surveillance program at a broker. For example, the alerts generated by our NN DTW anomaly detection model could be further analyzed using the finely tuned rules from existing expert systems in order to identify the most pertinent alerts that a surveillance analyst should review first. While we focus on insider trading, our framework is sufficiently generic to allow the detection of other forms of illegal trading activities that are performed on an intraday basis in limit order driven financial markets by modifying the multivariate feature set used by the DTW algorithm.

Although we show that our model works particularly well for detecting suspected illegal insider trading activity in the real world, our study has limitations. First, we only evaluated our model for a sample of illegal insider trading cases in equity markets, since this is currently the most prevalent task within the industry. Future research may wish to consider how the methodology could be extended to detect cross-asset or cross-market illegal trading strategies. Second, we acknowledge that the ARMA(1,1) model used as one of the benchmarks in our study is a relatively simplistic statistical model. Third, we evaluated our model over a 21-day validation period for each of the 14 illegal insider trading cases that we study. Future work may wish to test our NN DTW model over a longer horizon, such as an one-year validation period. Fourth, we use a set of features derived from a review of the financial market microstructure literature that describe the trading behaviour of individual broker accounts. Future research may wish to explore the performance of our approach for different feature sets.

In addition, future research could extend our model in the following ways. First, the warping window width parameter in the NN DTW algorithm could be set using a data driven approach. One possibility here is to use transfer learning. Alternatively recent work has recast DTW as an optimization problem which allows for a more general estimation procedure (see, e.g., Deriso and Boyd, 2019). Second, an ensemble model could be created by using popular feature bagging and

subsampling techniques that explore subspaces of the data to further improve detection performance. Finally, future research could consider how the contextual outlier detection approach of Golmohammadi et al. (2014) and the EDM approach of Amiram et al. (2020) could be applied to detect illegal trading activity based upon individual time series of broker transactions and if these approaches could be combined with our NN DTW model to improve the identification of illegal insider trading.

A Mechanics of Multivariate DTW

For the exposition which follows, let $\mathbf{A}$ and $\mathbf{B}$ denote two multivariate sequences of p features with m observations, that is $\mathbf{A}$ and $\mathbf{B}$ are $m \times p$ data matrices. We refer to the multivariate sequence $\mathbf{A}$ as the query sequence which is drawn from the testing data and to the multivariate sequence $\mathbf{B}$ as the candidate sequence which is drawn from the training data. We wish to define a mapping between the observations in $\mathbf{A}$ with respect to the observations in $\mathbf{B}$ by warping $\mathbf{A}$. It is assumed that all features have been standardized to zero mean and unit variance so the solution is scale-invariant. DTW begins by initializing an $m \times m$ cost matrix $\mathbf{C}$ containing the Euclidean distances between all possible combinations of row vectors $\{\mathbf{a}_i, i = 1 \dots, m\}$ and $\{\mathbf{b}_j, j = 1 \dots, m\}$ from $\mathbf{A}$ and $\mathbf{B}$ respectively,

$$\mathbf{C}_{ij} = \|\mathbf{a}_i - \mathbf{b}_j\|_2, \quad i, j = 1, \dots, m, \quad (4)$$

where $\|\cdot\|_2$ denotes the Euclidean norm. A contiguous warping path containing pairs of indices $\phi = \{\phi_l : \phi_l = (i, j)_l, l = 1, \dots, L\}$, is traced through the cost matrix to define the alignment between observations. Each observation $\mathbf{a}_i$ may be mapped to more than one $\mathbf{b}_j$, and vice versa. The alignment cost associated with a warping path, and therefore the distance between the two input sequences, is the sum of Euclidean distances along the warping path,

$$c_p = \sum_{l=1}^L \mathbf{C}_{\phi_l} \quad (5)$$

The optimal warping path produces the alignment of lowest cost, subject to a number of constraints. Formally, we can write the optimal warping path search problem as

$$\min_{\phi} \sum_{(i,j) \in \phi} C_{ij} \quad (6)$$

$$\text{Subject to: } \phi_1 = (1, 1) \quad (7)$$

$$\phi_L = (m, m) \quad (8)$$

$$\phi_l - \phi_{l-1} \in \{(1, 1), (0, 1), (1, 0)\} \quad (9)$$

$$\phi_l = \begin{cases} (i + \varphi, i) & \text{if } i > j + \varphi \\ (j, j + \varphi) & \text{if } j > i + \varphi \\ (i, j) & \text{otherwise} \end{cases} \quad (10)$$

The first three constraints (7), (8) and (9) state that the warping path must consider all points in both sequences, must be continuous with a step size of one and that no backward matching is allowed

(the slope of the warping path must be non-negative at all points). The fourth constraint (10), referred to within the literature as the *Warping Window Width constraint*, effectively restricts the cost matrix, and the credible region of the optimal warping path, to be a band matrix with identical upper and lower bandwidths which are equal to the constant φ . Intuitively, this constraint limits how far forward or backward in time an observation can be warped, to ensure the solution remains realistic. Restricting the search space of the optimal warping path also reduces the complexity of the DTW algorithm. For synchronous data, one obtains the Euclidean mapping between two sequences as a special case of DTW where the warping path lies along the diagonal of the cost matrix.

The optimal warping path is the solution to a dynamic programming problem. This solution is obtained by constructing a cumulative cost matrix, denoted by $\mathbf{C}^*$. The initial condition is

$$\mathbf{C}_{i,j}^* = \begin{cases} \infty, & \text{if } i = 1 \text{ or } j = 1 \\ 0, & \text{if } i = j = 1 \end{cases} \quad (11)$$

The recurrence relation used to obtain the solution is

$$\mathbf{C}_{i,j}^* = \mathbf{C}_{i,j} + \min \begin{cases} \mathbf{C}_{i-1,j}^* \\ \mathbf{C}_{i,j-1}^* \\ \mathbf{C}_{i-1,j-1}^* \end{cases} \quad \text{for } i, j = 2, \dots, m \quad (12)$$

Once the cumulative cost matrix $\mathbf{C}^*$ has been recursively filled the DTW distance between the two input sequences is

$$DTW(\mathbf{A}, \mathbf{B}) = \sqrt{\mathbf{C}_{m,m}^*} \quad (13)$$

B Complexity Reduction Techniques

At each iteration of the nearest neighbor search for a query broker account sequence, we store the best-so-far (BSF) DTW distance in memory. The BSF distance is defined as the current smallest DTW distance across all previously searched candidate broker account sequences in the training dataset. Lower bound functions rely on this BSF distance to prune candidate broker account sequences before computing the full DTW distance. A lower bound function takes as input, the query and candidate sequences and returns a scalar value that is guaranteed to be smaller than the true DTW distance. A pragmatic lower bound must be faster to compute than the true DTW distance and must closely approximate the true DTW distance, a property referred to as the "tightness" of the lower bound. The three lower bounds used in this study are implemented in a cascading fashion in order of complexity to determine if the current candidate training sequence can be pruned from the nearest neighbor search. We describe these bounds in ascending order of complexity.

The first lower bound, referred to as the modified *LB_Kim* lower bound (Rakthanmanon et al., 2013), recognizes that the sum of the Euclidean distance between the first pair of observations and the Euclidean distance between the last pair of observations is always a component of the total DTW distance, as per constraint (7) and (8). The modified *LB_Kim* lower bound is

$$LB_Kim(\mathbf{A}, \mathbf{B}) = \|\mathbf{a}_1 - \mathbf{b}_1\|_2 + \|\mathbf{a}_m - \mathbf{b}_m\|_2 \quad (14)$$

If in any iteration of the nearest neighbor search, *LB_Kim* exceeds the BSF distance then we may abandon the current candidate sequence comparison and proceed to the next iteration since the current candidate sequence cannot possibly be the nearest neighbor.

If *LB_Kim* does not exceed the BSF distance we compute a second tighter but computationally more expensive lower bound, known as *LB_Keogh* (Keogh and Ratanamahatana, 2005). The *LB_Keogh* lower bound builds an envelope around the query sequence based upon the size of the *Warping Window Width constraint* parameter φ . The *LB_Keogh* lower bound is

$$LB_Keogh(\mathbf{B}, \mathbf{U}, \mathbf{L}) = \sqrt{\sum_{q=1}^m \sum_{k=1}^p \begin{cases} (\mathbf{B}_{kq} - \mathbf{U}_{kq})^2 & \text{if } \mathbf{B}_{kq} > \mathbf{U}_{kq} \\ (\mathbf{B}_{kq} - \mathbf{L}_{kq})^2 & \text{if } \mathbf{B}_{kq} < \mathbf{L}_{kq} \\ 0 & \text{otherwise} \end{cases}} \quad (15)$$

where

$$\mathbf{U}_{kq} = \text{Max}(\mathbf{A}_{k-\varphi:k+\varphi,q}) \quad (16)$$

$$\mathbf{L}_{kq} = \text{Min}(\mathbf{A}_{k-\varphi:k+\varphi,q}) \quad (17)$$

are the upper and lower envelope sequences with a width dependent on the size of φ . Since the query sequence is fixed during the nearest neighbor search the upper and lower envelope sequences only need to be computed once in the first iteration and can be re-used for all other iterations. If the *LB_Keogh* lower bound exceeds the BSF distance then we may abandon the current candidate sequence comparison and proceed to the next iteration of the nearest neighbor search.

Finally, if both *LB_Kim* and *LB_Keogh* do not exceed the BSF distance a third lower bound is computed, referred to as *LB_Improved* (Lemire, 2009). The *LB_Improved* lower bound uses the information from the *LB_Keogh* lower bound to compute $\mathbf{B}'$, the projection of the candidate sequence $\mathbf{B}$ on the upper and lower envelopes

$$\mathbf{B}'_{kq} = \begin{cases} \mathbf{U}_{kq} & \text{if } \mathbf{B}_{kq} > \mathbf{U}_{kq} \\ \mathbf{B}_{kq} & \text{if } \mathbf{L}_{kq} < \mathbf{B}_{kq} < \mathbf{U}_{kq} \\ \mathbf{L}_{kq} & \text{if } \mathbf{B}_{kq} < \mathbf{L}_{kq} \end{cases}$$

Expression (16) and (17) are reused to compute $\mathbf{U}'_{kq}$ and $\mathbf{L}'_{kq}$ by replacing $\mathbf{A}$ with $\mathbf{B}'$. The *LB_Improved* lower bound is then,

$$LB_Improved = LB_Keogh(\mathbf{B}, \mathbf{U}, \mathbf{L}) + LB_Keogh(\mathbf{A}, \mathbf{U}', \mathbf{L}')$$

If all three aforementioned lower bounds are smaller than the BSF distance we proceed by implementing two early abandoning strategies during the computation of the DTW distance (Rakthanmanon et al., 2013). Early abandoning specifies conditions under which it is possible to stop the DTW distance computation early if the current candidate sequence cannot be the nearest neighbor. Both strategies make use of the recurrence relation (12), to access a partial value of the full DTW distance in the i^{th} iteration, which is $\min\{\mathbf{C}_{i1}^*, \dots, \mathbf{C}_{im}^*\}$. The first early abandoning strategy states that if the partial DTW distance exceeds the BSF distance the current candidate sequence comparison can be abandoned and we may proceed to the next iteration of the nearest neighbor search.

The second early abandoning strategy uses the sum of the partial DTW distance and the

remaining portion of the *LB_Keogh* lower bound, which is

$$\min \{ \mathbf{C}_{i1}^*, \dots, \mathbf{C}_{im}^* \} + LB_Keogh(\mathbf{A}_{i:m,1:m}, \mathbf{U}_{i:m,1:m}, \mathbf{L}_{i:m,1:m})$$

If this quantity exceeds the BSF we can abandon the current candidate sequence comparison and proceed to the next iteration of the nearest neighbor search.

Algorithm 1 presents pseudo code for the NN DTW anomaly scoring function. The nearest neighbor search begins by initializing the best-so-far DTW distance (*BSFdistance*) to infinity. We then search through all broker account sequences in the training dataset to find the nearest neighbor. Lines 3 to 9 perform the lower bound function checks implemented in cascading order of computational complexity. Lines 10 to 13 compute the true DTW distance and check if the DTW distance calculation was early abandoned, in which case the distance score is set to the placeholder value of infinity. The BSF distance is updated if the DTW distance score is smaller than the prevailing BSF distance in the current iteration.

Algorithm 1: NN DTW Anomaly Scoring Function

Input : Training dataset $\mathbf{X}$, Query Sequence $\mathbf{A}$
Output : Scalar Anomaly Score for the Query Sequence
Initialize: *BSFdistance* = ∞

```

1 for  $i = 1, \dots, |\mathbf{X}|$  do
2    $\mathbf{B} = \mathbf{X}_i$ 
3   if  $LB\_Kim(\mathbf{A}, \mathbf{B}) < BSFdistance$  then
4     if  $i = 1$  then
5       Construct upper and lower envelope sequences  $\mathbf{U}, \mathbf{L}$  (See (16) and (17))
6     if  $LB\_Keogh(\mathbf{B}, \mathbf{U}, \mathbf{L}) < BSFdistance = 0$  then
7       Construct projection sequence  $\mathbf{B}'$ 
8       Construct upper and lower envelope sequences  $\mathbf{U}', \mathbf{L}'$  around  $\mathbf{B}'$ 
9       if  $LB\_Keogh(\mathbf{B}, \mathbf{U}, \mathbf{L}) + LB\_Keogh(\mathbf{A}, \mathbf{U}', \mathbf{L}') < BSFdistance$  then
10        if  $DTW(\mathbf{A}, \mathbf{B})$  is early abandoned then
11           $DTW\_distance\_score = \infty$ 
12        else
13           $DTW\_distance\_score = DTW(\mathbf{A}, \mathbf{B})$ 
14        else
15           $DTW\_distance\_score = \infty$ 
16      else
17         $DTW\_distance\_score = \infty$ 
18    else
19       $DTW\_distance\_score = \infty$ 
20    if  $DTW\_distance\_score < BSFdistance$  then
21       $BSFdistance = DTW\_distance\_score$ 
22  $anomaly\_score = BSFdistance$ 

```

C Peaks-Over-Threshold Extreme Value Theory

Let F denote the common but unknown distribution of the anomaly scores. Exceedances of the anomaly scores above a high initial threshold value are the set $Z = \{\vartheta_{id} - u | \vartheta_{id} > u, i = 1 \dots N_d : d \in I_1\}$. The Pickands-Balkema-de Haan theorem (Pickands, 1975; Balkema and De Haan, 1974) states that the distribution of the exceedances converges to the generalized Pareto family of distributions for a suitably large threshold value,

$$\lim_{u \rightarrow F^+} \sup_{0 < z < \vartheta^{(X)+} - u} |Pr(Z < z) - G_{\xi, \sigma}(z)| = 0 \quad (18)$$

where F^+ is the right endpoint of F , $\vartheta^{(X)+}$ is the largest training sample anomaly score and $G_{\xi, \sigma}$ is the generalized Pareto distribution function with shape parameter ξ and scale parameter σ .

The generalized Pareto distribution function is

$$G_{\xi, \sigma}(z) = \begin{cases} 1 - \left(1 + \frac{\xi z}{\sigma}\right)^{-1/\xi}, & \text{for } \xi \neq 0, z > 0 \\ 1 - \exp\left(-\frac{z}{\sigma}\right), & \text{for } \xi = 0, z > 0 \end{cases} \quad (19)$$

When $\xi > 0$, the generalized Pareto distribution is heavy tailed. The shape and scale parameters are estimated via Maximum Likelihood Estimation where the log-likelihood function is given in Coles (2001). Let $\hat{\xi}$ and $\hat{\sigma}$ denote the parameter estimates.

Assuming that the tail of F , the distribution of training sample anomaly scores, begins at u , then for some $\eta > u$, the following inequality holds for observations in the tail:

$$F(\eta) = 1 - (1 - F(u))(1 - F_u(\eta - u)), \quad \eta > u \quad (20)$$

where F_u is the distribution of the excess loss beyond u . The POT method constructs an estimate for $F(\eta)$. The term $(1 - F(u))$ is approximated non-parametrically by the proportion of observations in the tail region above u , which is N_u/N , where $N_u = |Z|$ is the cardinality of Z and $N = |\vartheta^{(X)}|$ is the cardinality of $\vartheta^{(X)}$. The distribution of the excess loss beyond $F_u(\eta - u)$ is generalized Pareto by the Pickands-Balkema-de Haan theorem. Hence, the POT tail estimator is

$$\hat{F}(\eta) = \hat{F}_Z(z) = 1 - \frac{N_u}{N} \left(1 + \frac{\xi z}{\sigma}\right)^{-1/\xi}, \quad \eta > u \quad (21)$$

An estimator of the extreme α -quantile can be obtained by inverting the POT tail estimator

$\hat{F}_Z$ as follows:

$$\hat{Q}^{(\alpha)} = \hat{F}_Z^{-1}(\alpha) \equiv u + \frac{\hat{\sigma}}{\hat{\xi}} \left[\left(\frac{N}{N_u} (1 - \alpha) \right)^{-\hat{\xi}} - 1 \right], \quad \alpha > F(u) \quad (22)$$

To derive an estimator of the CTE we use the fact that $E[\vartheta_{id} - u | \vartheta_{id} > u, i = 1 \dots N_d : d \in I_1] = \sigma/(1 - \xi)$ (for $\xi < 1$) and that for some $u' > u$ the excedances $\{\vartheta_{id} - u' | \vartheta_{id} > u'\}$ also follow a generalized Pareto distribution with an identical shape parameter ξ but a transformed scale parameter $\sigma' = \sigma + \xi(u' - u)$ (Coles, 2001).

D Time Series Cross Validation Details

In the first iteration the first set of training sample anomaly scores, denoted by $\vartheta_1^{(X)}$, is constructed by computing the NN DTW distance for each broker account sequence with respect to all other broker account sequences in the first 6-month training dataset. The remaining steps in the first iteration of the time series cross-validation process, and the steps required in all subsequent iterations are summarized as follows. For $d = 1, \dots, 21$,

1. Compute the d^{th} anomaly score threshold, denoted $\tau_d^{(\alpha)}$, by estimating the parameters of the generalized Pareto distribution based upon $\vartheta_d^{(X)}$, as outlined in Section 5.
2. Construct the d^{th} set of testing sample anomaly scores, denoted $\vartheta_d^{(Y)}$, by computing the NN DTW distance for each broker account sequence in the d^{th} one-day testing dataset with respect to all broker account sequences in the d^{th} six-month training dataset.
3. Construct the d^{th} class label vector $\hat{\mathbf{y}}_{(d)} = -\text{sign}(\vartheta_d^{(Y)} - \tau_d^{(\alpha)})$.
4. Construct the $(d+1)^{th}$ six-month training sample by appending sequences classified as normal in the d^{th} testing sample to the end of the d^{th} six-month training sample. Remove all sequences for the first day in $(d+1)^{th}$ six-month training sample.
5. Construct the $(d+1)^{th}$ set of training sample anomaly scores by appending anomaly scores for sequences classified as normal in the d^{th} testing sample to the end of the d^{th} set of training sample anomaly scores. Remove all sequences for the first day in $(d+1)^{th}$ set of training sample anomaly scores.

The set of training sample anomaly scores needs to be built from scratch only once in the first iteration of the cross-validation procedure. In all subsequent iterations, the set of training sample anomaly scores is updated based upon the class labels assigned in the previous iteration. In practice, since the first step is computationally intensive, the construction of $\vartheta_1^{(X)}$ could be performed offline, before the implementation of the detection model.

Table 1: Trading Activity Statistics for the Suspected Insider Account

The table reports descriptive statistics of trading activity for the suspected illegal insider account in each case. The variable *First Trade Date* is the number of days prior to the information release date that the insider started trading. The variable *Insider Volume* is expressed in millions of U.S. dollars.

	First Trade date	No. Insider Trades	Average No. Insider trades per day active	Insider Volume
Case 1	4	254	127	2.70
Case 2	9	105	26	0.50
Case 3	1	13	13	1.53
Case 4	7	1070	178	7.70
Case 5	10	2230	278	90.00
Case 6	2	84	84	1.71
Case 7	2	583	583	10.66
Case 8	7	196	196	3.43
Case 9	2	264	132	1.70
Case 10	2	183	91.5	4.31
Case 11	1	134	134	2.46
Case 12	1	511	511	19.96
Case 13	2	37	37	1.52
Case 14	1	238	238	1.14
Mean	3.64	421.57	187.75	10.66
Total		5902		149.31

Table 2: Abnormal Returns Around Information Release Dates

The table reports Abnormal return information for each alleged illegal insider trading case in the sample. The variable $CAR(-10, -1)$ denotes the cumulative abnormal return over the 10-day pre information release period. The variable $AR(0)$ denotes the information release day abnormal return. Abnormal returns are expressed as a percentage. The model used to compute abnormal returns and the bootstrap procedure used to establish statistical significance is described in (Dubow and Monteiro, 2006). ***, ** and * represent statistical significance at the 1%, 5% and 10% significance levels, respectively.

	CAR(-10,-1)	AR(0)
Case 1	1.281	-10.184
Case 2	-4.82	-40.796
Case 3	-5.552	-7.455
Case 4	-7.318	-10.473
Case 5	38.336***	20.187
Case 6	55.682	-6.646
Case 7	10.775	-7.224
Case 8	-6.983	-10.582
Case 9	1.629	-8.899
Case 20	2.787	-6.519
Case 11	-1.048	-5.174
Case 12	4.357	4.379
Case 13	-6.609**	-5.063
Case 14	-0.578	-7.817

Table 3: Summary Statistics for Model Features

The table reports summary statistics for each of the features used in our illegal insider trading detection model, computed for the initial six-month training dataset and 10-day validation dataset. Mathematical definitions of each feature are presented in the online supplementary materials file. T-statistics are reported for differences in the mean of each feature between the training dataset and validation sample. With the exception of the feature *Participation rate*, all trade volume statistics are reported in 000's of shares. The features *Immediate Price Impact*, *Adverse Selection Spread Component* and *Realized Volatility* have been multiplied by 100 for exposition. ***, ** and * represent statistical significance at the 1%, 5% and 10% significance levels, respectively.

	(i) Training Sample					(ii) Validation Sample					
	Mean	Median	Std	Min	Max	Mean	Median	Std	Min	Max	Difference
Panel A: Trade Volume Features											
Volume	7.199	2.000	22.884	0.100	1127.000	7.894	2.000	20.602	0.100	700.000	3.914***
Short Sell Volume	3.380	0.000	17.186	0.000	1127.000	3.871	0.000	15.684	0.000	411.000	3.677***
Residual Trade Volume	0.126	-2.461	22.264	-94.827	1117.946	0.289	-2.600	19.822	-76.803	665.629	0.945
Moving Window Volume	329.624	52.800	1624.612	0.000	26184.000	216.472	65.400	359.969	0.100	3344.000	-9.551***
Participation Rate	0.052	0.020	0.087	0.000	1.000	0.072	0.027	0.116	0.000	1.000	26.878***
Panel B: Illiquidity Features											
Immediate Price Impact	0.022	0.000	0.125	-4.577	3.611	0.022	0.000	0.127	-1.667	2.102	-0.331
Adverse Selection Spread Component	-0.004	0.014	1.086	-14.297	18.934	0.075	0.030	1.233	-11.128	10.526	8.955***
Adjusted Trade Duration	0.154	0.021	0.288	0.000	1.000	0.152	0.026	0.281	0.000	1.000	-0.884
Panel C: Volatility Feature											
Realized Volatility	0.025	0.010	0.063	0.000	1.522	0.026	0.011	0.042	0.000	0.476	2.507**

Table 4: Area Under the Curve Statistics for the 1-NN DTW and ARMA(1,1) Illegal Insider Trading detection model. The table reports Area Under the Receiver Operating Characteristic (AUROC) curve statistics and Area Under the Precision Recall (AUPR) curve statistics for the NN DTW illegal insider trading detection model and the three detection algorithms from the ARMA(1,1) benchmark model of (Park and Lee, 2010) for all 14 cases. Differences in mean area under the curve statistics between the NN DTW model and each of the three ARMA(1,1) model detection algorithms are presented in the final row. Statistical significance of each difference is established using a Wilcoxon Signed Ranks Test. ***, ** and * represent statistical significance at the 1%, 5% and 10% significance levels, respectively.

	(i) Receiver Operating Characteristic Curve						(ii) Precision Recall Curve					
	NN DTW			ARMA(1,1) Algorithm 1			ARMA(1,1) Algorithm 2			ARMA(1,1) Algorithm 3		
	Area	Mean	Std	Area	Mean	Std	Area	Mean	Std	Area	Mean	Std
Case 1	1.000	0.550	0.579	0.647	0.658	0.666	0.686	0.690	0.701	0.712	0.723	0.734
Case 2	1.000	0.500	0.833	0.762	0.850	0.938	0.907	0.918	0.929	0.940	0.951	0.962
Case 3	0.997	0.538	0.720	0.935	0.501	0.325	0.011	0.022	0.033	0.044	0.055	0.066
Case 4	0.855	0.536	0.573	0.772	0.578	0.559	0.587	0.598	0.609	0.620	0.631	0.642
Case 5	0.907	0.529	0.514	0.609	0.943	0.934	0.934	0.945	0.956	0.967	0.978	0.989
Case 6	1.000	0.515	0.535	0.180	0.501	0.501	0.001	0.012	0.023	0.034	0.045	0.056
Case 7	1.000	0.439	0.568	0.303	0.885	0.924	0.816	0.827	0.838	0.849	0.860	0.871
Case 8	0.944	0.528	0.590	0.383	0.608	0.586	0.134	0.145	0.156	0.167	0.178	0.189
Case 9	0.992	0.551	0.618	0.742	0.633	0.649	0.690	0.701	0.712	0.723	0.734	0.745
Case 10	0.992	0.545	0.496	0.355	0.513	0.443	0.132	0.143	0.154	0.165	0.176	0.187
Case 11	1.000	0.620	0.647	0.244	0.632	0.629	0.107	0.118	0.129	0.140	0.151	0.162
Case 12	0.984	0.545	0.619	0.535	0.511	0.491	0.199	0.210	0.221	0.232	0.243	0.254
Case 13	1.000	0.709	0.701	0.888	0.513	0.512	0.531	0.542	0.553	0.564	0.575	0.586
Case 14	0.935	0.581	0.876	0.250	0.979	0.995	0.936	0.947	0.958	0.969	0.980	0.991
Average	0.972	0.549***	0.634***	0.543***	0.665**	0.666**	0.477***	0.488	0.499	0.510	0.521	0.532

Table 5: Information Retrieval Statistics for the NN DTW and ARMA(1,1) Illegal Insider Trading Detection models. The table reports Precision, Recall and F1 scores for the NN DTW illegal insider trading detection model and for the three detection algorithms from the ARMA(1,1) benchmark model. Differences in mean information retrieval statistics between the NN DTW model and each of the three ARMA(1,1) model detection algorithms are presented in the final row. Statistical significance of each difference is established using a Wilcoxon Signed Ranks Test. ***, ** and * represent statistical significance at the 1%, 5% and 10% significance levels, respectively.

	(i) NN DTW			(ii) ARMA(1,1) Algorithm 1			(iii) ARMA(1,1) Algorithm 2			(iv) ARMA(1,1) Algorithm 3		
	Precision	Recall	F1	Precision	Recall	F1	Precision	Recall	F1	Precision	Recall	F1
Case 1	1.000	1.000	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Case 2	1.000	1.000	1.000	0.500	0.429	0.462	0.000	0.000	0.000	0.600	0.429	0.500
Case 3	0.022	1.000	0.042	0.002	1.000	0.003	0.000	0.000	0.000	0.001	1.000	0.003
Case 4	0.365	0.718	0.484	0.047	0.217	0.077	0.047	0.058	0.052	0.047	0.217	0.077
Case 5	0.910	0.581	0.709	0.834	0.513	0.635	0.852	0.124	0.217	0.834	0.514	0.636
Case 6	0.150	1.000	0.261	0.007	1.000	0.015	0.000	0.000	0.000	0.007	1.000	0.014
Case 7	0.739	1.000	0.850	0.942	0.539	0.686	0.067	0.002	0.004	0.931	0.531	0.676
Case 8	0.601	0.562	0.581	0.247	0.964	0.393	0.088	0.089	0.088	0.239	0.964	0.383
Case 9	0.974	1.000	0.987	0.010	0.013	0.011	0.000	0.000	0.000	0.009	0.013	0.010
Case 10	0.377	1.000	0.547	0.036	0.766	0.069	0.026	0.125	0.042	0.036	0.766	0.068
Case 11	0.717	1.000	0.835	0.307	1.000	0.470	0.023	0.029	0.026	0.282	1.000	0.440
Case 12	0.651	0.914	0.760	0.098	0.860	0.175	0.046	0.177	0.073	0.096	0.860	0.173
Case 13	1.000	1.000	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Case 14	0.916	0.870	0.892	0.978	0.964	0.971	0.846	0.158	0.267	0.964	0.971	0.968
Average	0.673	0.903	0.711	0.286***	0.59**	0.283***	0.142***	0.054***	0.055***	0.289***	0.59**	0.282***

Table 6: Area Under the Curve Statistics for the NN DTW, EGMM, iForest and OCSVM Illegal Insider Trading Detection models.

The table reports Area Under the Receiver Operating Characteristic (AUROC) curve statistics and Area Under the Precision Recall (AUPR) curve statistics for the NN DTW illegal insider trading detection model and the Ensemble Gaussian Mixture (EGMM), isolation Forest (iForest) and One-Class Support Vector Machine (OCSVM) detection models. Differences in mean area under the curve statistics between the NN DTW models and each of the EGMM, iForest and OCSVM models are presented in the final row. Statistical significance of each difference is established using a Wilcoxon Signed Ranks Test. ***, ** and * represent statistical significance at the 1%, 5% and 10% significance levels, respectively.

	(i) Receiver Operating Characteristic Curve				(ii) Precision Recall Curve			
	NN DTW	EGMM	iForest	OCSVM	NN DTW	EGMM	iForest	OCSVM
Case 1	1.000	0.726	0.833	0.806	1.000	0.596	0.493	0.366
Case 2	1.000	1.000	0.580	1.000	1.000	1.000	0.639	1.000
Case 3	0.997	0.976	0.960	0.915	0.325	0.158	0.303	0.031
Case 4	0.855	0.842	0.763	0.619	0.537	0.579	0.365	0.154
Case 5	0.907	0.877	0.743	0.761	0.912	0.871	0.721	0.793
Case 6	1.000	0.654	0.774	0.759	1.000	0.068	0.303	0.082
Case 7	1.000	0.932	0.897	0.914	1.000	0.927	0.819	0.872
Case 8	0.944	0.676	0.754	0.708	0.736	0.333	0.361	0.239
Case 9	0.992	0.684	0.743	0.662	0.899	0.372	0.437	0.275
Case 10	0.992	0.787	0.868	0.879	0.848	0.090	0.201	0.361
Case 11	1.000	0.547	0.798	0.860	1.000	0.097	0.231	0.530
Case 12	0.984	0.943	0.910	0.871	0.756	0.463	0.316	0.174
Case 13	1.000	0.994	0.995	0.888	1.000	0.977	0.919	0.114
Case 14	0.935	0.859	0.564	0.629	0.982	0.910	0.744	0.758
Average	0.972	0.821***	0.799***	0.805***	0.857	0.532***	0.489***	0.41***

Table 7: Information Retrieval Statistics for the NN DTW, EGMM, iForest and OCSVM Illegal Insider Trading Detection models.

The table reports precision, recall and F1 scores for the NN DTW illegal insider trading detection model and for the Ensemble Gaussian Mixture (EGMM), isolation Forest (iForest) and One-Class Support Vector Machine (OCSVM) models. Differences in mean information retrieval statistics between the NN DTW model and each of the EGMM, iForest and OCSVM models are presented in the final row. Statistical significance of each difference is established using a Wilcoxon Signed Ranks Test. ***, ** and * represent statistical significance at the 1%, 5% and 10% significance levels, respectively.

	(i) NN DTW			(ii) EGMM			(iii) iForest			(iv) OCSVM		
	Precision	Recall	F1	Precision	Recall	F1	Precision	Recall	F1	Precision	Recall	F1
Case 1	1.000	1.000	1.000	0.200	0.780	0.318	0.469	0.327	0.385	0.260	0.925	0.406
Case 2	1.000	1.000	1.000	0.856	1.000	0.922	0.595	0.143	0.230	0.606	1.000	0.755
Case 3	0.022	1.000	0.042	0.013	1.000	0.026	0.028	0.846	0.054	0.007	1.000	0.014
Case 4	0.365	0.718	0.484	0.252	0.950	0.399	0.299	0.641	0.408	0.167	0.883	0.282
Case 5	0.910	0.581	0.709	0.680	0.942	0.790	0.726	0.542	0.621	0.567	0.898	0.695
Case 6	0.150	1.000	0.261	0.053	1.000	0.100	0.186	0.452	0.264	0.061	0.964	0.114
Case 7	0.739	1.000	0.850	0.590	0.971	0.734	0.803	0.770	0.786	0.423	0.991	0.593
Case 8	0.601	0.562	0.581	0.343	0.613	0.440	0.413	0.341	0.374	0.225	0.931	0.363
Case 9	0.974	1.000	0.987	0.455	0.595	0.516	0.567	0.481	0.520	0.268	0.913	0.414
Case 10	0.377	1.000	0.547	0.100	0.978	0.181	0.196	0.596	0.295	0.090	0.940	0.164
Case 11	0.717	1.000	0.835	0.122	0.455	0.193	0.246	0.351	0.289	0.134	0.970	0.236
Case 12	0.651	0.914	0.760	0.192	0.928	0.318	0.310	0.753	0.439	0.103	0.982	0.186
Case 13	1.000	1.000	1.000	0.112	1.000	0.201	0.353	0.973	0.518	0.051	1.000	0.098
Case 14	0.916	0.870	0.892	0.922	0.996	0.958	0.750	0.025	0.049	0.769	0.882	0.822
Average	0.673	0.903	0.711	0.349***	0.872	0.435***	0.424***	0.517***	0.374***	0.267***	0.949	0.367***

Table 8: Simulation Results

The table reports the classification rates from the simulation exercise described in Section 6.6. The simulation design parameter *n.modified* is the number of observations that are modified in the randomly chosen broker account sequence. The simulation design parameter *warping* is the degree of warping that is added to the randomly chosen broker account sequence.

	TNR	FPR	TPR	FNR	TNR	FPR	TPR	FNR
	n.modified = 100, warping = 30%				n.modified = 100, warping = 50%			
NN DTW	1.000	0.000	0.999	0.001	1.000	0.000	1.000	0.000
OCSVM	1.000	0.000	0.127	0.873	1.000	0.000	0.119	0.881
EGMM	1.000	0.000	0.152	0.848	1.000	0.000	0.103	0.897
iForest	1.000	0.000	0.898	0.102	1.000	0.000	0.907	0.094
	n.modified = 90, warping = 30%				n.modified = 90, warping = 50%			
NN DTW	0.999	0.001	0.999	0.001	1.000	0.000	0.997	0.003
OCSVM	0.979	0.021	0.149	0.851	0.989	0.011	0.140	0.860
EGMM	0.976	0.024	0.171	0.830	0.988	0.012	0.117	0.883
iForest	0.934	0.066	0.904	0.096	0.931	0.069	0.903	0.097
	n.modified = 70, warping = 30%				n.modified = 70, warping = 50%			
NNN DTW	0.938	0.062	0.998	0.002	0.928	0.072	0.999	0.001
OCSVM	0.937	0.063	0.180	0.820	0.968	0.032	0.181	0.819
EGMM	0.928	0.072	0.200	0.800	0.965	0.035	0.141	0.859
iForest	0.797	0.203	0.906	0.094	0.788	0.212	0.907	0.093
	n.modified = 50, warping = 30%				n.modified = 50, warping = 50%			
NN DTW	0.828	0.172	0.999	0.001	0.843	0.157	1.000	0.000
OCSVM	0.893	0.107	0.231	0.769	0.941	0.059	0.223	0.777
EGMM	0.877	0.123	0.239	0.761	0.936	0.064	0.173	0.827
iForest	0.645	0.355	0.909	0.091	0.631	0.369	0.912	0.088

References

- Ahmad, S., A. Lavin, S. Purdy, and Z. Agha (2017). Unsupervised real-time anomaly detection for streaming data. *Neurocomputing* 262, 134–147.
- Aitken, M., D. Cumming, and F. Zhan (2015). Exchange trading rules, surveillance and suspected insider trading. *Journal of Corporate Finance* 34, 311–330.
- Aitken, M. J., F. H. d. B. Harris, and S. Ji (2015). A worldwide examination of exchange market quality: Greater integrity increases market efficiency. *Journal of Business Ethics* 132(1), 147–170.
- Amiram, D., E. Lyandres, and D. Rabetti (2020). Competition and product quality: Fake trading on crypto exchanges. *Available at SSRN 3745617*.
- ASIC (2019). Review of Australian Equity Market Cleanliness. Technical report, Australian Securities and Investments Commission.
- Ausubel, L. M. (1990). Insider trading in a rational expectations economy. *The American Economic Review* 80(5), 1022–1041.
- Axelsson, S. (2000). The base-rate fallacy and the difficulty of intrusion detection. *ACM Transactions on Information and System Security (TISSEC)* 3(3), 186–205.
- Back, K. and S. Baruch (2004). Information in securities markets: Kyle meets glosten and milgrom. *Econometrica* 72(2), 433–465.
- Bader, B., J. Yan, and X. Zhang (2018). Automated threshold selection for extreme value analysis via ordered goodness-of-fit tests with adjustment for false discovery rate. *The Annals of Applied Statistics* 12(1), 310–329.
- Bagnall, A., J. Lines, A. Bostrom, J. Large, and E. Keogh (2017). The great time series classification bake off: a review and experimental evaluation of recent algorithmic advances. *Data Mining and Knowledge Discovery* 31(3), 606–660.
- Balkema, A. A. and L. De Haan (1974). Residual life time at great age. *The Annals of Probability* 2(5), 792–804.
- Bolton, R. J. and D. J. Hand (2002). Statistical fraud detection: A review. *Statistical Science* 17(3), 235–255.

- Chandola, V., A. Banerjee, and V. Kumar (2009). Anomaly detection: A survey. *ACM Computing Surveys (CSUR)* 41(3), 15.
- Cline, B. N. and V. V. Posylnaya (2019). Illegal insider trading: Commission and sec detection. *Journal of Corporate Finance* 58, 247–269.
- Coles, S. (2001). *An introduction to statistical modeling of extreme values*, Volume 208. Springer.
- Comerton-Forde, C. and T. J. Putniņš (2011). Measuring closing price manipulation. *Journal of Financial Intermediation* 20(2), 135–158.
- Cong, L. W., X. Li, K. Tang, and Y. Yang (2021). Crypto wash trading. *arXiv preprint arXiv:2108.10984*.
- Cumming, D., S. Johan, and D. Li (2011). Exchange trading rules and stock market liquidity. *Journal of Financial Economics* 99(3), 651–671.
- Dau, H. A., D. F. Silva, F. Petitjean, G. Forestier, A. Bagnall, A. Mueen, and E. Keogh (2018). Optimizing dynamic time warping’s window width for time series data mining applications. *Data Mining and Knowledge Discovery* 32(4), 1074–1120.
- DeMarzo, P. M., M. J. Fishman, and K. M. Hagerty (1998). The optimal enforcement of insider trading regulations. *Journal of Political Economy* 106(3), 602–632.
- Demšar, J. (2006). Statistical comparisons of classifiers over multiple data sets. *Journal of Machine Learning Research* 7(1), 1–30.
- Deng, S., C. Wang, M. Wang, and Z. Sun (2019). A gradient boosting decision tree approach for insider trading identification: An empirical model evaluation of china stock market. *Applied Soft Computing* 83, 105652.
- Deriso, D. and S. Boyd (2019). A general optimization framework for dynamic time warping. *arXiv preprint arXiv:1905.12893*.
- Derrig, R. A. (2002). Insurance fraud. *Journal of Risk and Insurance* 69(3), 271–287.
- Diaz, D., B. Theodoulidis, and P. Sampaio (2011). Analysis of stock market manipulations using knowledge discovery techniques applied to intraday trade prices. *Expert Systems with Applications* 38(10), 12757–12771.

- Ding, H., G. Trajcevski, P. Scheuermann, X. Wang, and E. Keogh (2008). Querying and mining of time series data: experimental comparison of representations and distance measures. *Proceedings of the VLDB Endowment* 1(2), 1542–1552.
- Dubow, B. and N. B. Monteiro (2006). Measuring market cleanliness. *Available at SSRN 1019999*.
- Easley, D. and M. O’hara (1987). Price, trade size, and information in securities markets. *Journal of Financial Economics* 19(1), 69–90.
- Emmott, A., S. Das, T. Dietterich, A. Fern, and W.-K. Wong (2015). A meta-analysis of the anomaly detection problem. *arXiv preprint arXiv:1503.01158*.
- Fernandes, N. and M. A. Ferreira (2009). Insider trading laws and stock price informativeness. *The Review of Financial Studies* 22(5), 1845–1887.
- Glosten, L. R. and P. R. Milgrom (1985). Bid, ask and transaction prices in a specialist market with heterogeneously informed traders. *Journal of Financial Economics* 14(1), 71–100.
- Golmohammadi, K., O. R. Zaiane, and D. Díaz (2014). Detecting stock market manipulation using supervised learning algorithms. In *2014 International Conference on Data Science and Advanced Analytics (DSAA)*, pp. 435–441. IEEE.
- G’Sell, M. G., S. Wager, A. Chouldechova, and R. Tibshirani (2016). Sequential selection procedures and false discovery rate control. *Journal of the Royal Statistical society: Series B (Statistical Methodology)* 78(2), 423–444.
- Hasbrouck, J. (1991). Measuring the information content of stock trades. *The Journal of Finance* 46(1), 179–207.
- Hautsch, N. and M. Podolskij (2013). Preaveraging-based estimation of quadratic variation in the presence of noise and jumps: Theory, implementation, and empirical evidence. *Journal of Business & Economic Statistics* 31(2), 165–183.
- He, H. and J. Wang (1995). Differential information and dynamic behavior of stock trading volume. *The Review of Financial Studies* 8(4), 919–972.
- Islam, S. R., S. K. Ghafoor, and W. Eberle (2018). Mining illegal insider trading of stocks: a proactive approach. In *2018 IEEE International Conference on Big Data (Big Data)*, pp. 1397–1406. IEEE.

- Kacperczyk, M. and E. S. Pagnotta (2019). Chasing private information. *The Review of Financial Studies* 32(12), 4997–5047.
- Kaniel, R. and H. Liu (2006). So what orders do informed traders use? *The Journal of Business* 79(4), 1867–1913.
- Keogh, E. and C. A. Ratanamahatana (2005). Exact indexing of dynamic time warping. *Knowledge and Information Systems* 7(3), 358–386.
- Kurbalija, V., M. Radovanović, Z. Geler, and M. Ivanović (2014). The influence of global constraints on similarity measures for time-series databases. *Knowledge-Based Systems* 56, 49–67.
- Lee, E. J., K. S. Eom, and K. S. Park (2013). Microstructure-based manipulation: Strategic behavior and performance of spoofing traders. *Journal of Financial Markets* 16(2), 227–252.
- Leland, H. E. (1992). Insider trading: Should it be prohibited? *Journal of Political Economy* 100(4), 859–887.
- Lemire, D. (2009). Faster retrieval with a two-pass dynamic-time-warping lower bound. *Pattern Recognition* 42(9), 2169–2180.
- Lim, M. and R. Coggins (2005). The immediate price impact of trades on the Australian stock exchange. *Quantitative Finance* 5(4), 365–377.
- Liu, F. T., K. M. Ting, and Z.-H. Zhou (2012). Isolation-based anomaly detection. *ACM Transactions on Knowledge Discovery from Data (TKDD)* 6(1), 1–39.
- Manove, M. (1989). The harm from insider trading and informed speculation. *The Quarterly Journal of Economics* 104(4), 823–845.
- Meulbroek, L. K. (1992). An empirical analysis of illegal insider trading. *The Journal of Finance* 47(5), 1661–1699.
- Ngai, E. W., Y. Hu, Y. H. Wong, Y. Chen, and X. Sun (2011). The application of data mining techniques in financial fraud detection: A classification framework and an academic review of literature. *Decision Support Systems* 50(3), 559–569.
- Olmo, J., K. Pilbeam, and W. Pouliot (2011). Detecting the presence of insider trading via structural break tests. *Journal of Banking & Finance* 35(11), 2820–2828.
- Park, Y. S. and J. Lee (2010). Detecting insider trading: The theory and validation in Korea exchange. *Journal of Banking & Finance* 34(9), 2110–2120.

- Patton, A. J. and K. Sheppard (2009). Optimal combinations of realised volatility estimators. *International Journal of Forecasting* 25(2), 218–238.
- Pedregosa, F., G. Varoquaux, A. Gramfort, V. Michel, B. Thirion, O. Grisel, M. Blondel, P. Prettenhofer, R. Weiss, V. Dubourg, et al. (2011). Scikit-learn: Machine learning in python. *Journal of Machine Learning Research* 12(85), 2825–2830.
- Pickands, J. (1975). Statistical inference using extreme order statistics. *The Annals of Statistics* 3(1), 119–131.
- Putniņš, T. J. (2012). Market manipulation: A survey. *Journal of Economic Surveys* 26(5), 952–967.
- Rakthanmanon, T., B. Campana, A. Mueen, G. Batista, B. Westover, Q. Zhu, J. Zakaria, and E. Keogh (2013). Addressing big data time series: Mining trillions of time series subsequences under dynamic time warping. *ACM Transactions on Knowledge Discovery from Data (TKDD)* 7(3), 1–31.
- Sakoe, H. and S. Chiba (1978). Dynamic programming algorithm optimization for spoken word recognition. *IEEE Transactions on Acoustics, Speech, and Signal Processing* 26(1), 43–49.
- Schölkopf, B., J. C. Platt, J. Shawe-Taylor, A. J. Smola, and R. C. Williamson (2001). Estimating the support of a high-dimensional distribution. *Neural Computation* 13(7), 1443–1471.
- Shkilko, A. (2019). Insider trading under the microscope. *Manuscript, Wilfrid Laurier University*.
- Shokoohi-Yekta, M., B. Hu, H. Jin, J. Wang, and E. Keogh (2017). Generalizing dtw to the multi-dimensional case requires an adaptive approach. *Data Mining and Knowledge Discovery* 31(1), 1–31.
- Silva, D. F., R. Giusti, E. Keogh, and G. E. Batista (2018). Speeding up similarity search under dynamic time warping by pruning unpromising alignments. *Data Mining and Knowledge Discovery* 32(4), 988–1016.
- Spathis, C., M. Doumpos, and C. Zopounidis (2002). Detecting falsified financial statements: a comparative study using multicriteria analysis and multivariate statistical techniques. *European Accounting Review* 11(3), 509–535.